

Mellin–Lerch Transforms and Dilation Identities for Stieltjes Correlations

Polylogarithms, generalized harmonic sums, and exact finite parts

Research continuation prepared for the ProveIt project
Analytic derivations and computational audit with OpenAI assistance

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Abstract

We develop three exact identity calculi that continue the ProveIt manuscript and its five latest incoming reports. First, a Mellin transform of a polylogarithm against $(1+x)^{-N}$ is reduced, for every positive integer N and every complex spectral order, to a finite sum of differences of Lerch transcendents. The differences have cancelling positive-axis jumps. All integer Mellin parameters in the convergence strip are removable resonances, and every logarithmic moment there reduces to finitely many ordinary polylogarithms. At unit argument, spectral differentiation gives integral identities for log-gamma, polygamma, and generalized Stieltjes constants, with explicit harmonic coefficients at zeta poles. Second, we give a two-variable digamma reflection formula for shifted generalized harmonic sums and an exact coefficient procedure. Its unshifted parity specializations have published antecedents and are credited accordingly. Third, we resolve the incoming question about integer dilation and finite-part normalization. Explicit local polynomials describe the difference between distributional pullback and constant-term subtraction in the ambient coordinate. Unequal dilations reduce to one shift governed by their greatest common divisor; the resulting polygamma product formula involves only Hurwitz zeta and its first spectral derivative, with a necessary logarithm of the least common multiple. All assertions include analytic proofs. The accompanying computations check finite algebra and compare independent integral or series representations; they do not replace those proofs. The Gaussian S_6 and S_8 evaluations remain conjectural.

Repository snapshot. `fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed`.

This article is a proposed research addition. Its analytic proofs are given in full; finite symbolic checks and numerical diagnostics are supplied separately.

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1 Scope, source audit, and mathematical status

The canonical manuscript already contains extensive polylogarithmic reduction, gamma arithmetic, Hurwitz-parameter differentiation, and Stieltjes antiderivative theory. The latest incoming reports add periodic Hurwitz convolution, shifted correlations, their spectral derivatives, and contact terms under differentiation. A useful continuation must therefore distinguish an additional identity from a repackaging of an existing result.

We inspected the manuscript text and status files and all five archives in the incoming directory at commit `fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed` [1, 8, 9, 10, 11, 12]. The source manifest records the retrieved bytes and their hashes. The audit was directed at the identity questions relevant to this article; it is not a claim to have re-proved every theorem in the manuscript.

Source strand	Consequence for the present work
Manuscript integration and differentiation chapters	The elementary-symmetric harmonic derivative tower and Hurwitz-jet antiderivatives are established starting points.
Manuscript Gaussian harmonic sums	The recorded S_4 evaluation has an exact proof. The residual certificates for S_6 and S_8 do not prove equality.
Incoming shifted Hurwitz jets	The explicit question about dilation, traces, and subtraction scale is answered in Section 5.
Other four incoming packages	Their convolution, resonance, and correlation identities constrain the novelty and normalization claims here.
Published Euler-sum literature	The unshifted parity reductions in Section 4 are consistent with Xu–Wang’s prior formulas.

1.1 What is proved here

The main additions are an explicit finite transform formula for arbitrary denominator order, its complete treatment at integer Mellin resonances, a shifted harmonic reflection generator, and a dilation theorem that retains the local finite-part terms. These are identity results. Estimates appear only when needed to justify convergence, parameter differentiation, or continuation.

The Mellin formula is derived from the classical Fermi integral, Euler’s beta integral, and elementary partial fractions. The higher-pole reduction and its integer-parameter formulas organize those ingredients into a uniform calculation. The two-shift harmonic theorem packages an infinite family into one identity of analytic functions. The dilation result closes a question explicitly proposed in the incoming material, with a precise choice of subtraction coordinate.

None of these descriptions asserts global literature priority for every specialization. The standard special-function conventions follow the DLMF [2, 3, 4, 5]. The relevant Euler-sum antecedent is Xu and Wang [6]; the underlying Hurwitz product-integral method goes back at least to the work of Espinosa and Moll [7]. The article supplies its own proofs of every formula used as a new addition to the repository.

1.2 Conventions

For $|z| < 1$ and $\Re q > 0$,

$$\mathrm{Li}_s(z) = \sum_{n \geq 1} \frac{z^n}{n^s}, \quad \Phi(z, s, q) = \sum_{n \geq 0} \frac{z^n}{(n+q)^s}.$$

Thus $\text{Li}_s(z) = z\Phi(z, s, 1)$. Powers of $n + q$ use the principal logarithm on the right half-plane; more generally principal logarithms are used unless a continuation is specified. We write $\psi(q) = \Gamma'(q)/\Gamma(q)$. The gamma logarithm is the analytic $\log \Gamma(q)$ on $\Re q > 0$, normalized by $\log \Gamma(1) = 0$.

Spectral derivatives are denoted by square brackets:

$$\text{Li}_s^{[m]}(z) = \partial_s^m \text{Li}_s(z), \quad \zeta^{[m]}(s, q) = \partial_s^m \zeta(s, q).$$

An ordinary superscript in $\psi^{(r)}(q)$ or $\gamma_m^{(r)}(q)$ means differentiation in q . The normalization of the generalized Stieltjes constants is

$$\zeta(1+u, q) = \frac{1}{u} + \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(q) u^m, \quad \gamma_0(q) = -\psi(q). \quad (1.1)$$

We write $\gamma_m = \gamma_m(1)$ and $\gamma = \gamma_0$. Generalized harmonic numbers are

$$H_n^{(r)} = \sum_{j=1}^n j^{-r}, \quad H_0^{(r)} = 0, \quad H_n = H_n^{(1)}.$$

The rising factorial is $(s)_r = s(s+1) \cdots (s+r-1)$, with $(s)_0 = 1$. Coefficient extraction, such as $[u^m]F(u)$, always refers to an analytic germ or to a specified formal power series. A removable product such as $u\zeta(1+u)$ is evaluated as a whole.

There are two different regularization operations later in the article. Distributional pullback transports an already extended distribution. Ambient-coordinate finite part instead extends the dilated pointwise function by deleting divergent terms in the original integration coordinate. Their difference is explicit and generally nonzero. Throughout the Mellin and harmonic sections, the integrals are ordinary convergent integrals and no finite-part convention is involved.

2 A finite Mellin–Lerch calculus

2.1 The analytic difference that removes the Lerch cut

Put $\mathcal{S} = \mathbb{C} \setminus (-\infty, 0]$. For $z \in \mathcal{S}$ we use $\log z$ with imaginary part in $(-\pi, \pi)$. The natural expression in the transform is

$$\mathcal{L}(s, q; z) = z^q \Phi(z, s, q) - \text{Li}_s(z), \quad \Re q > 0. \quad (2.1)$$

Initially this is defined by the two convergent series for $0 < z < 1$. The individual summands have the usual positive-axis continuation cut; their difference does not.

Lemma 2.1 (Cancellation and the unit-argument germ). *The expression (2.1) extends to a holomorphic function of $z \in \mathcal{S}$, holomorphic in $\Re q > 0$, and entire in s . Its value at $z = 1$ is*

$$\mathcal{L}(s, q; 1) = \zeta(s, q) - \zeta(s), \quad (2.2)$$

where the pole at $s = 1$ cancels. For $z > 1$, either consistent lateral continuation of the two terms in (2.1) gives this same function. Locally at $z = 1$, with $\mu = \log z$,

$$\mathcal{L}(s, q; e^\mu) = \sum_{k \geq 0} \frac{\zeta(s-k, q) - \zeta(s-k)}{k!} \mu^k, \quad |\mu| < 2\pi, \quad (2.3)$$

understood on the local logarithm sheet. The series and its fixed-order parameter derivatives converge locally uniformly.

Proof. For $\Re s > 0$ the Lerch integral writes the first term as

$$\frac{z^q}{\Gamma(s)} \int_0^\infty \frac{t^{s-1} e^{-qt}}{1 - ze^{-t}} dt.$$

For $z > 1$, the residue at $t = \log z$ is $(\log z)^{s-1}/\Gamma(s)$, independent of q . It is therefore the same as the residue for $q = 1$, which gives $\text{Li}_s(z)$. The two lateral jumps cancel. This argument first applies away from exceptional spectral orders and extends to them analytically.

At $\mu = 0$ the classical Lerch expansion [3, §25.14] has the form

$$e^{q\mu} \Phi(e^\mu, s, q) = \Gamma(1-s)(-\mu)^{s-1} + \sum_{k \geq 0} \zeta(s-k, q) \frac{\mu^k}{k!}.$$

Subtract its $q = 1$ version. The nonanalytic term is independent of q and cancels exactly, giving (2.3). At integer s , subtract first and continue in s ; each coefficient is a difference of two Hurwitz zeta functions with the same residue. This proves regularity at $z = 1$ and (2.2). Local uniform convergence follows from the convergent Lerch expansion on smaller μ -disks and Cauchy estimates on compact parameter sets. The series definition gives entire dependence on s for $0 < z < 1$, and continuation preserves it. There are no remaining obstructions in $\mathcal{S}$: the positive-axis jumps have cancelled, $z = 1$ is removable, and $z = 0$ is excluded by the chosen domain. Equivalently, the integral representation in Theorem 2.3 below independently supplies this continuation when $0 < \Re q < 2$, after which the finite Lerch shift identity extends it to every $\Re q > 0$. $\square$

Remark 2.2. The local logarithm qualification in (2.3) matters. It is a power series in μ , not a license to replace $\log z$ by an arbitrary logarithm while keeping the other branches fixed. In computations at $z > 1$, taking inconsistent lateral values before subtraction introduces a spurious imaginary term.

2.2 The master transform at every denominator order

For an integer $N \geq 1$ define

$$J_N(a, s; z) = \int_0^\infty \frac{x^{a-1} \text{Li}_s(-zx)}{(1+x)^N} dx. \quad (2.4)$$

Let the rational polynomials $q_{N,j}(a)$ be determined by

$$Q_{N,a}(X) := \binom{X+a-1}{N-1} = \frac{1}{(N-1)!} \prod_{r=1}^{N-1} (X+a-r) = \sum_{j=0}^{N-1} q_{N,j}(a) X^j. \quad (2.5)$$

For $N = 1$ the product is 1.

Theorem 2.3 (Finite Mellin–Lerch transform). *Let $N \geq 1$, $z \in \mathcal{S}$, $s \in \mathbb{C}$, and*

$$-1 < \Re a < N. \quad (2.6)$$

Then (2.4) is an ordinary convergent integral, holomorphic in these parameters and entire in s . For noninteger a ,

$$J_N(a, s; z) = \frac{(-1)^N \pi}{\sin \pi a} \sum_{j=0}^{N-1} q_{N,j}(a) \mathcal{L}(s-j, N-a; z). \quad (2.7)$$

All the apparent singularities at $a = 0, 1, \dots, N-1$ are removable. The same formula, continued as a whole, holds there. The strip (2.6) is the maximal common strip for all spectral orders: already $s = 0, z = 1$ forces both endpoints.

Proof. We first establish uniform convergence and spectral holomorphy, including negative spectral orders. Fix compact sets of s and $z \in \mathcal{S}$. Choose an integer $r \geq 0$ such that $\Re(s+r) > 0$ throughout a neighborhood of the spectral compact set. The Fermi integral gives

$$\text{Li}_{s+r}(-zx) = -\frac{1}{\Gamma(s+r)} \int_0^\infty t^{s+r-1} \frac{zxe^{-t}}{1+zxe^{-t}} dt.$$

Apply $(x\partial_x)^r$ to obtain $\text{Li}_s(-zx)$. The differentiated rational kernel is bounded by a constant times $\min(xe^{-t}, 1)$, uniformly in these compact sets: the ray of zxe^{-t} remains a positive angular distance from the negative real axis, and its denominator is bounded away from zero relative to $1 + |zxe^{-t}|$. Integrating this bound, splitting at $t = \log x$ when $x > 1$, shows

$$\text{Li}_s(-zx) = O(x) \quad (x \downarrow 0), \quad \text{Li}_s(-zx) = O((1 + \log x)^M) \quad (x \rightarrow \infty)$$

for some finite M , uniformly on the chosen compacts. These estimates give an integrable majorant in every smaller strip in (2.6). They justify holomorphy of (2.4). They also justify arbitrary fixed powers of $\log x$, and spectral differentiation follows either directly or from Cauchy estimates. For $s = 0, z = 1$, $\text{Li}_0(-x) = -x/(1+x)$ shows the sharpness of the common strip.

For $N = 1$, take initially $0 < z < 1$, $0 < \Re a < 1$, and $\Re s > 0$. Fubini's theorem applies to the Fermi integral. If $b > 0$, partial fractions and the beta integral give

$$\int_0^\infty \frac{x^a}{(1+x)(x+b)} dx = \frac{\pi}{\sin \pi a} \frac{b^a - 1}{b - 1}. \quad (2.8)$$

At $b = 1$ the right side is interpreted by continuity. One can prove (2.8) first for $-1 < \Re a < 0$, where the two partial fractions are separately integrable, and then continue to $-1 < \Re a < 1$, where their difference is integrable. This avoids separating divergent integrals.

Substitute $b = e^t/z$ into (2.8). The result is

$$\begin{aligned} J_1(a, s; z) &= -\frac{\pi}{\Gamma(s) \sin \pi a} \int_0^\infty t^{s-1} \frac{z^{1-a} e^{at} - z}{e^t - z} dt \\ &= -\frac{\pi}{\sin \pi a} [z^{1-a} \Phi(z, s, 1-a) - \text{Li}_s(z)]. \end{aligned}$$

This proves (2.7) for $N = 1$ on an open set.

Let E be the spectral backward shift, $(Ef)(s) = f(s-1)$. Integration of the derivative of $x^a(1+x)^{-N} \text{Li}_s(-zx)$, with vanishing endpoint terms, gives

$$J_{N+1}(a, s; z) = \frac{(N-a)J_N(a, s; z) - J_N(a, s-1; z)}{N}. \quad (2.9)$$

Consequently, in a common strip,

$$J_N = \frac{1}{(N-1)!} \prod_{r=1}^{N-1} (r-a-E) J_1.$$

Applying this polynomial in E to the Lerch series annihilates its first $N-1$ terms, because

$$Q_{N,a}(n+1-a) = \binom{n}{N-1} = 0 \quad (0 \leq n < N-1). \quad (2.10)$$

Shift the remaining index by $N-1$. The Lerch parameter becomes $N-a$, and the power z^{1-a} becomes z^{N-a} . Since

$$Q_{N,a}(n+N-a) = \binom{n+N-1}{N-1},$$

the resulting expression is exactly (2.7). The parameter $N - a$ has positive real part on the entire strip (2.6). Holomorphic continuation in a and s , followed by Lemma 2.1 in z , proves the identity throughout the asserted domain. The original integral is holomorphic at every integer in that strip, so the remaining cosecant poles are removable. The next theorem gives their values without relying on an unspecified limit. $\square$

The first two cases illustrate the finite character of the result:

$$J_1(a, s; z) = -\pi \csc(\pi a) \mathcal{L}(s, 1 - a; z), \quad (2.11)$$

$$J_2(a, s; z) = \pi \csc(\pi a) [\mathcal{L}(s - 1, 2 - a; z) + (a - 1) \mathcal{L}(s, 2 - a; z)]. \quad (2.12)$$

For general N , only N Lerch differences occur. The shift to $N - a$ is essential: retaining $1 - a$ term by term would leave the safe Hurwitz half-plane when $\Re a > 1$ and obscure the cancellations in (2.10).

2.3 Integer Mellin parameters and complete logarithmic moments

Define the entire spectral function

$$F_s(z) = s \operatorname{Li}_{s+1}(z) - \log z \operatorname{Li}_s(z), \quad (2.13)$$

with the same consistent continuation across $z > 1$. At $z = 1$ this means

$$F_s(1) = s\zeta(s+1), \quad F_0(1) = 1. \quad (2.14)$$

The last value is a removable value, not the undefined product $0 \cdot \zeta(1)$.

Theorem 2.4 (Every integer resonance). *For $0 \leq m \leq N - 1$,*

$$\boxed{J_N(m, s; z) = (-1)^{N+m} \sum_{j=0}^{N-1} q_{N,j}(m) F_{s-j}(z).} \quad (2.15)$$

More strongly, for sufficiently small δ ,

$$J_N(m + \delta, s; z) = (-1)^{N+m-1} \sum_{j=0}^{N-1} q_{N,j}(m + \delta) J_1(\delta, s - j; z). \quad (2.16)$$

Thus every $\log^h x$ moment at an integer Mellin parameter is a finite combination of ordinary polylogarithms, powers of $\log z$, and even zeta values, with coefficients polynomial in s and rational coefficients.

Proof. Write $L = N - m$. The finite Lerch shift formula is

$$\mathcal{L}(s, L - \delta; z) = \mathcal{L}(s, 1 - \delta; z) - \sum_{k=1}^{L-1} \frac{z^{k-\delta}}{(k - \delta)^s}.$$

Multiply its shifted spectral versions by $q_{N,j}(m + \delta)$ and sum in j . Every term in the finite sum vanishes identically in δ , since

$$Q_{N,m+\delta}(k - \delta) = \binom{k + m - 1}{N - 1} = 0 \quad (1 \leq k < L).$$

Using $\sin \pi(m + \delta) = (-1)^m \sin \pi \delta$ proves (2.16). The first Taylor coefficient in

$$\mathcal{L}(s, 1 - \delta; z) = e^{-\delta \log z} \sum_{r \geq 0} \frac{(s)_r}{r!} \operatorname{Li}_{s+r}(z) \delta^r - \operatorname{Li}_s(z)$$

is $F_s(z)$. This series calculation is first made for $0 < z < 1$, where every coefficient is finite, and then continued as a combined analytic germ. At $z = 1$ and spectral resonances the removable products are evaluated only after combination. Hence $J_1(0, s; z) = -F_s(z)$, proving (2.15). Further coefficient extraction proves the last assertion, made explicit below. $\square$

Let

$$\frac{\pi t}{\sin \pi t} = \sum_{j \geq 0} e_{2j} t^{2j}, \quad e_0 = 1, \quad e_{2j} = 2(1 - 2^{1-2j})\zeta(2j) \quad (j \geq 1),$$

and define, for $r \geq 1$,

$$d_r(s, z) = \sum_{v=0}^r \frac{(-\log z)^{r-v} (s)_v}{(r-v)! v!} \text{Li}_{s+v}(z). \quad (2.17)$$

Then

$$\mathcal{M}_h(s; z) := \int_0^\infty \frac{\log^h x \text{Li}_s(-zx)}{x(1+x)} dx = -h! \sum_{j=0}^{\lfloor h/2 \rfloor} e_{2j} d_{h+1-2j}(s, z). \quad (2.18)$$

Combining this with (2.16) gives the completely finite rule

$$\begin{aligned} & \int_0^\infty \frac{x^{m-1} \log^h x \text{Li}_s(-zx)}{(1+x)^N} dx \\ &= (-1)^{N+m-1} \sum_{j=0}^{N-1} \sum_{v=0}^h \binom{h}{v} q_{N,j}^{(v)}(m) \mathcal{M}_{h-v}(s-j; z). \end{aligned} \quad (2.19)$$

Terms with $v > N-1$ vanish automatically. All parameter differentiations under these integrals are justified by the uniform majorants in the proof of Theorem 2.3.

At $z = 1$, the especially short version is

$$\boxed{\mathcal{M}_h(s; 1) = -h! \sum_{j=0}^{\lfloor h/2 \rfloor} e_{2j} \frac{(s)_{h+1-2j}}{(h+1-2j)!} \zeta(s+h+1-2j).} \quad (2.20)$$

For example,

$$\begin{aligned} \mathcal{M}_0(s; 1) &= -s\zeta(s+1), \\ \mathcal{M}_1(s; 1) &= -\frac{s(s+1)}{2}\zeta(s+2), \\ \mathcal{M}_2(s; 1) &= -\frac{s(s+1)(s+2)}{3}\zeta(s+3) - 2\zeta(2)s\zeta(s+1), \\ \mathcal{M}_3(s; 1) &= -\frac{(s)_4}{4}\zeta(s+4) - 3\zeta(2)(s)_2\zeta(s+2). \end{aligned}$$

In particular, the $s = 1, h = 2$ case gives the ordinary integral

$$\int_0^\infty \frac{\log^2 x \log(1+x)}{x(1+x)} dx = 7\zeta(4).$$

Example 2.5 (Two finite higher-pole reductions). The $N = 3$ integer rows are

$$\begin{aligned} J_3(1, s; z) &= \frac{1}{2}(F_{s-2}(z) - F_{s-1}(z)), \\ J_3(2, s; z) &= -\frac{1}{2}(F_{s-2}(z) + F_{s-1}(z)). \end{aligned}$$

At $z = 1, s = 1$ they equal $-1/4$ and $-3/4$. The necessary input is $F_0(1) = 1$; simply dropping a factor proportional to $s - 1$ would give wrong answers.

2.4 Finite algebra, negative orders, and rational kernels

The coefficient table in (2.15) is nondegenerate as a finite polynomial transformation.

Proposition 2.6 (Invertible integer table). *For $0 \leq m, j \leq N - 1$,*

$$\det(q_{N,j}(m)) = \frac{(-1)^{N(N-1)/2}}{\prod_{j=0}^{N-1} j!}. \quad (2.21)$$

Consequently the N Mellin integrals in (2.15) determine, by rational linear algebra, the functions

$$F_s(z), \quad F_{s-1}(z), \quad \dots, \quad F_{s-N+1}(z).$$

Proof. Replace the rows indexed by m by their successive forward differences at $m = 0$. The row operation has determinant one. Pascal's identity gives

$$\Delta_m^k Q_{N,m}(X)|_{m=0} = \binom{X-1}{N-1-k}.$$

These rows have descending degrees $N - 1 - k$ and leading coefficients $1/(N - 1 - k)!$. Their coefficient matrix is triangular after reversing the column order. The reversal contributes $(-1)^{N(N-1)/2}$, proving (2.21). $\square$

This is a statement about a finite coefficient matrix. It makes no claim about arithmetic independence of values at one spectral order.

For a separate exact check at negative spectral orders, let $\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \}$ denote a Stirling number of the second kind. The rational expression for the negative-order polylogarithm yields

$$J_N(a, -r; 1) = \sum_{k=0}^r (-1)^{k+1} k! \left\{ \begin{smallmatrix} r+1 \\ k+1 \end{smallmatrix} \right\} B(a+k+1, N-a), \quad r \geq 0. \quad (2.22)$$

Indeed,

$$\text{Li}_{-r}(-x) = \sum_{k=0}^r (-1)^{k+1} k! \left\{ \begin{smallmatrix} r+1 \\ k+1 \end{smallmatrix} \right\} \frac{x^{k+1}}{(1+x)^{k+1}},$$

and each term integrates to the beta factor in (2.22). Differentiating this finite beta expression gives an independent gamma/polygamma evaluation of the logarithmic moments at those orders. It checks the spectral-pole cancellations without evaluating a singular zeta product numerically.

Corollary 2.7 (Proper rational kernels with negative real poles). *Let*

$$R(x) = \sum_{\nu=1}^M \sum_{k=1}^{N_\nu} \frac{c_{\nu,k}}{(x+b_\nu)^k}, \quad b_\nu > 0.$$

For $-1 < \Re a < 1$,

$$\int_0^\infty x^{a-1} R(x) \text{Li}_s(-zx) dx = \sum_{\nu,k} c_{\nu,k} b_\nu^{a-k} J_k(a, s; z b_\nu). \quad (2.23)$$

In particular every logarithmic moment at $a = 0$ reduces finitely by (2.19); spectral derivatives reduce in the same way. If cancellation in R gives a larger convergence strip, (2.23) continues there as a combined expression.

Proof. Partial fractions are finite, so the common strip permits termwise integration. Substitute $x = b_\nu y$ in each term. Logarithmic moments follow by differentiating in a , including the elementary factors b_ν^{a-k} . Analytic continuation applies to the complete finite sum on any larger strip where the original integral is holomorphic. $\square$

This corollary is an exact reduction procedure: denominator multiplicity, Mellin resonance, logarithmic moments, and spectral derivatives are handled by finite operations. It does not assert a comparable reduction for products of two arbitrary polylogarithms; that extension is a separate problem.

3 Gamma, polygamma, and Stieltjes identities

The same transform has two useful spectral centers. At $s = 0$, the first derivative of Hurwitz zeta gives $\log \Gamma$. At $s = 1$, the pole-free Hurwitz difference gives the generalized Stieltjes constants. Integer Mellin parameters then supply finite expressions in ordinary zeta jets. The resulting identities are convergent integral formulas, not finite parts.

3.1 A log-gamma kernel and every polygamma derivative

For $j \geq 0$ and $0 < \Re q < 2$, set

$$\mathcal{G}_j(q) = \int_0^\infty \frac{x^{-q} \log^j x}{1+x} \text{Li}_0^{[1]}(-x) dx. \quad (3.1)$$

Theorem 3.1 (Log-gamma and polygamma integral identities). *On $0 < \Re q < 2$,*

$$\boxed{\log \Gamma(q) = -\frac{\sin \pi q}{\pi} \mathcal{G}_0(q).} \quad (3.2)$$

For every integer $r \geq 1$,

$$\psi^{(r-1)}(q) = -\sum_{j=0}^r \binom{r}{j} (-1)^j \pi^{r-j-1} \sin\left(\pi q + \frac{(r-j)\pi}{2}\right) \mathcal{G}_j(q). \quad (3.3)$$

All integrals are ordinary convergent integrals. In particular, $q = 1$ is included without assigning a singular value to either side of (3.2).

Proof. Set $a = 1 - q$ and $z = 1$ in (2.11), and differentiate spectrally at $s = 0$. The classical identity

$$\zeta^{[1]}(0, q) = \log \Gamma(q) - \frac{1}{2} \log(2\pi)$$

gives

$$\mathcal{G}_0(q) = -\pi \csc(\pi q) \log \Gamma(q)$$

away from $q = 1$. Multiplying by the sine and continuing gives (3.2) everywhere in the strip. Differentiation in q gives $\partial_q^j \mathcal{G}_0(q) = (-1)^j \mathcal{G}_j(q)$. Apply the product rule and the formula for successive derivatives of $\sin \pi q$ to obtain (3.3). Theorem 2.3 supplies the uniform majorants for every derivative. $\square$

For example,

$$\psi(q) = -\cos(\pi q) \mathcal{G}_0(q) + \frac{\sin \pi q}{\pi} \mathcal{G}_1(q).$$

At the half point, (3.2) becomes

$$\int_0^\infty \frac{\text{Li}_0^{[1]}(-t^2)}{1+t^2} dt = -\frac{\pi}{4} \log \pi. \quad (3.4)$$

Here $x = t^2$ converts $\mathcal{G}_0(1/2)$ to twice the displayed integral. For any two parameters a, b in the strip, subtraction also gives the gamma-ratio formula

$$\log \frac{\Gamma(b)}{\Gamma(a)} = -\frac{1}{\pi} \int_0^\infty \frac{\sin(\pi b)x^{-b} - \sin(\pi a)x^{-a}}{1+x} \text{Li}_0^{[1]}(-x) dx. \quad (3.5)$$

The left side means the difference of the analytic gamma logarithms fixed in Section 1.

3.2 Generalized Stieltjes constants without a divergent integral

Theorem 3.2 (Stieltjes difference transform). *For $m \geq 0$ and $0 < \Re q < 2$,*

$$\gamma_m(q) - \gamma_m = \frac{(-1)^{m+1} \sin \pi q}{\pi} \int_0^\infty \frac{x^{-q}}{1+x} \text{Li}_1^{[m]}(-x) dx. \quad (3.6)$$

For every integer $m \geq 1$,

$$\int_0^\infty \frac{\text{Li}_0^{[m]}(-x)}{x(1+x)} dx = (-1)^m m \gamma_{m-1}, \quad (3.7)$$

and equivalently

$$\int_0^\infty \frac{\text{Li}_1^{[m]}(-x)}{(1+x)^2} dx = (-1)^m m \gamma_{m-1}. \quad (3.8)$$

The zeroth-derivative values of both integrals in (3.7)–(3.8) are -1 .

Proof. At $s = 1$, the difference $\zeta(s, q) - \zeta(s)$ is entire and its m th derivative is $(-1)^m (\gamma_m(q) - \gamma_m)$. Apply this observation to (2.11) at $a = 1 - q$. This proves (3.6).

The integer resonance $J_1(0, s; 1) = -s\zeta(s+1)$ and the convergent expansion

$$s\zeta(s+1) = 1 + \sum_{n \geq 0} \frac{(-1)^n \gamma_n}{n!} s^{n+1} \quad (3.9)$$

give (3.7). The case $N = 2, m = 1$ of (2.15) is $J_2(1, s; 1) = -(s-1)\zeta(s)$. Differentiate at $s = 1$ to obtain (3.8). Alternatively, integration by parts directly shifts the polylogarithm order in the two integrals. All order derivatives commute with these ordinary integrals by Theorem 2.3. $\square$

At $q = 1/2$, differentiating the elementary identity $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$ gives

$$\gamma_1(1/2) - \gamma_1 = -2\gamma \log 2 - \log^2 2.$$

Thus a concrete first-order specialization of (3.6) is

$$\int_0^\infty \frac{\text{Li}_1^{[1]}(-t^2)}{1+t^2} dt = -\frac{\pi}{2} (2\gamma \log 2 + \log^2 2). \quad (3.10)$$

Equation (3.8) supplies, for instance,

$$\int_0^\infty \frac{\text{Li}_1^{[1]}(-x)}{(1+x)^2} dx = -\gamma, \quad \int_0^\infty \frac{\text{Li}_1^{[2]}(-x)}{(1+x)^2} dx = 2\gamma_1.$$

3.3 The exact harmonic coefficients at spectral resonances

The products in (2.20) are entire although their displayed zeta factor can have a pole. This is the point where finite generalized harmonic coefficients enter the Mellin calculus.

Proposition 3.3 (All derivatives of a cancelled zeta pole). *For an integer $r \geq 1$, put $s = 1 - r + u$. Then, as an analytic identity near $u = 0$,*

$$(s)_r \zeta(s+r) = (-1)^{r-1} (r-1)! \prod_{j=1}^{r-1} \left(1 - \frac{u}{j}\right) \left(1 + \sum_{n \geq 0} \frac{(-1)^n \gamma_n}{n!} u^{n+1}\right). \quad (3.11)$$

Consequently its m th spectral derivative at $s = 1 - r$ is

$$(-1)^{r-1} (r-1)! m! [u^m] \left\{ \prod_{j=1}^{r-1} \left(1 - \frac{u}{j}\right) \left(1 + \sum_{n \geq 0} \frac{(-1)^n \gamma_n}{n!} u^{n+1}\right) \right\}. \quad (3.12)$$

Only finitely many Stieltjes constants and harmonic coefficients enter a specified derivative.

Proof. Factor the one vanishing factor from the rising factorial:

$$(1 - r + u)_r = (-1)^{r-1} (r-1)! u \prod_{j=1}^{r-1} (1 - u/j).$$

Multiply by the Laurent series for $\zeta(1+u)$. This proves the germ and its derivative formula. $\square$

The polynomial coefficients are elementary symmetric functions in $1, 1/2, \dots, 1/(r-1)$. In terms of generalized harmonic numbers,

$$\prod_{j=1}^{r-1} (1 - u/j) = \exp \left(- \sum_{k \geq 1} \frac{H_{r-1}^{(k)}}{k} u^k \right).$$

The right side is used as a formal expansion to any required finite order. Its first coefficients are

$$1 - H_{r-1} u + \frac{H_{r-1}^2 - H_{r-1}^{(2)}}{2} u^2 - \frac{H_{r-1}^3 - 3H_{r-1} H_{r-1}^{(2)} + 2H_{r-1}^{(3)}}{6} u^3 + \dots$$

This is a finite coefficient mechanism, distinct from interpreting an undefined expression $0 \cdot \infty$ by numerical cancellation.

For example, since $\mathcal{M}_1(s; 1) = -\frac{1}{2}(s)_2 \zeta(s+2)$, expansion at $s = -1$ gives

$$\int_0^\infty \frac{\log x \operatorname{Li}_{-1}(-x)}{x(1+x)} dx = \frac{1}{2}, \quad (3.13)$$

$$\int_0^\infty \frac{\log x \operatorname{Li}_{-1}^{[1]}(-x)}{x(1+x)} dx = \frac{\gamma - 1}{2}, \quad (3.14)$$

$$\int_0^\infty \frac{\log x \operatorname{Li}_{-1}^{[2]}(-x)}{x(1+x)} dx = -\gamma - \gamma_1. \quad (3.15)$$

The first identity can also be checked using the rational function $\operatorname{Li}_{-1}(-x) = -x/(1+x)^2$ and a beta derivative. The spectral derivatives in the next two are different functions and should not be replaced by derivatives of that rational expression in x .

Higher denominator orders combine the same ingredients with the finite polynomial table. For example,

$$\int_0^\infty \frac{\operatorname{Li}_1^{[1]}(-x)}{(1+x)^3} dx = \frac{1}{4} \log(2\pi) - \frac{1}{4} - \frac{\gamma}{2}. \quad (3.16)$$

This follows by differentiating $J_3(1, s; 1) = \frac{1}{2}(F_{s-2}(1) - F_{s-1}(1))$ at $s = 1$, using $\zeta(0) = -1/2$ and $\zeta^{[1]}(0) = -\frac{1}{2} \log(2\pi)$.

3.4 Antiderivatives and the existing Hurwitz-jet ladder

The Mellin transform also produces a natural anchored antiderivative in its exponent parameter. This connects the new integral formulas to the antiderivative framework already present in the manuscript; the Hurwitz primitive itself is classical.

Proposition 3.4 (Anchored primitive). *For $-1 < \Re a < 1$ define*

$$\mathcal{P}_s(a) = \frac{\zeta(s-1, 1-a) - \zeta(s-1)}{s-1} - a\zeta(s), \quad (3.17)$$

where the complete expression is continued at exceptional s . It is entire in s , vanishes at $a = 0$, and satisfies

$$\partial_a \mathcal{P}_s(a) = \zeta(s, 1-a) - \zeta(s) = -\frac{\sin \pi a}{\pi} J_1(a, s; 1). \quad (3.18)$$

At the Stieltjes center,

$$\partial_s^m \mathcal{P}_s(a)|_{s=1} = \frac{\zeta^{[m+1]}(0, 1-a) - \zeta^{[m+1]}(0)}{m+1} - a(-1)^m \gamma_m. \quad (3.19)$$

In particular,

$$\mathcal{P}_1(a) = \log \Gamma(1-a) - \gamma a. \quad (3.20)$$

Proof. The Hurwitz parameter derivative $\partial_q \zeta(v, q) = -v\zeta(v+1, q)$ proves (3.18) away from the displayed singularities. The right side is entire in s , and its integral from 0 to a gives an entire continuation of (3.17). At $s = 1 + u$, the numerator has constant term $\zeta(0, 1-a) - \zeta(0) = a$. Its quotient contributes a/u , cancelling the pole of $a\zeta(1+u)$. Comparing the remaining coefficients gives (3.19), and the log-gamma identity gives (3.20). $\square$

For completeness, every positive integer primitive order r has a finite anchored expression:

$$\mathcal{P}_{s,r}(a) = \frac{\zeta(s-r, 1-a) - \sum_{j=0}^{r-1} \frac{(s-r)_j \zeta(s-r+j)}{j!} a^j}{(s-r)_r} - \frac{a^r}{r!} \zeta(s). \quad (3.21)$$

Away from the exceptional spectral parameters, r differentiations in a give (3.18); all lower initial derivatives vanish. Equivalently,

$$\mathcal{P}_{s,r}(a) = \frac{1}{(r-1)!} \int_0^a (a-v)^{r-1} [\zeta(s, 1-v) - \zeta(s)] dv.$$

This integral proves that all apparent spectral singularities in the combined expression (3.21) are removable. Taylor extraction at those singularities produces the complete homogeneous harmonic coefficients familiar from the manuscript's negative-polygamma ladder. The anchor fixes the integration polynomial, so no unrecorded integration constants enter.

Finally, argument differentiation and logarithmic integration commute with the Mellin operation:

$$z \partial_z J_N(a, s; z) = J_N(a, s-1; z), \quad \int \frac{J_N(a, s; z)}{z} dz = J_N(a, s+1; z) + C. \quad (3.22)$$

The primitive is local on a simply connected part of $\mathcal{S}$. This follows directly from the polylogarithm order ladder and the compact bounds proved earlier.

4 A two-shift reflection for generalized harmonic sums

The generalized harmonic order and the elementary-symmetric harmonic index in the repository are different parameters. We use new notation to keep them separate. Put

$$h_n^{(r)}(u) = \sum_{j=1}^n \frac{1}{(j-u)^r}, \quad Q_{p,r} = \sum_{n=1}^{\infty} \frac{(-1)^n H_n^{(r)}}{(2n+1)^p}, \quad p, r \geq 1.$$

Thus $h_n^{(r)}(0) = H_n^{(r)}$, and $Q_{p,1} = S_p$ in the manuscript. In contrast, the manuscript's $S_{p,r}$ uses $e_r(n) = [v^r] \prod_{j=1}^n (1 + v/j)$; it is not $Q_{p,r}$.

The unshifted even-weight evaluations belong to the established theory of linear Euler R -sums. In the normalization of Xu–Wang [6, Section 3.4, equation (20)],

$$R_{r,p}^{1,-1} = 2^p Q_{p,r}. \quad (4.1)$$

We derive a two-variable digamma resummation, its arbitrary mixed derivatives, and a concrete rational-shift evaluation. The scalar parity theorem is credited to that literature. The priority of the particular two-variable formulation below has not been established.

4.1 Ordinary convergence, including the first outer power

Define

$$A(z, u) = \sum_{n=1}^{\infty} \frac{(-1)^n h_n^{(1)}(u)}{2n+1-z}, \quad (4.2)$$

$$D(v) = \frac{1}{4} \left\{ \psi\left(\frac{v+3}{4}\right) - \psi\left(\frac{v+1}{4}\right) \right\}. \quad (4.3)$$

Initially, $D(v) = \int_0^1 x^v/(1+x^2) dx$ for $\Re v > -1$; the displayed expression supplies its meromorphic continuation. The first sum in (4.2) is an ordinary convergent alternating series.

Lemma 4.1 (Local uniform convergence). *For positive integers p, r , the series*

$$\sum_{n=1}^{\infty} \frac{(-1)^n h_n^{(r)}(u)}{(2n+1-z)^p}$$

converges locally uniformly on

$$\Omega_+ = (\mathbb{C} \setminus \{3, 5, 7, \dots\}) \times (\mathbb{C} \setminus \{1, 2, 3, \dots\}).$$

It is holomorphic there and can be differentiated term by term any finite number of times in z and u . It is absolutely convergent when $p \geq 2$.

Proof. Fix a compact subset of Ω_+ . Uniformly on that subset, $h_n^{(1)}(u) = O(1 + \log n)$, while $h_n^{(r)}(u) = O(1)$ for $r \geq 2$. For $f_n = h_n^{(r)}(u)(2n+1-z)^{-p}$, the exact difference is

$$\begin{aligned} f_{n+1} - f_n &= \frac{1}{(n+1-u)^r (2n+3-z)^p} \\ &\quad + h_n^{(r)}(u) \{ (2n+3-z)^{-p} - (2n+1-z)^{-p} \}. \end{aligned}$$

Consequently $f_n \rightarrow 0$ uniformly and $f_{n+1} - f_n = O(n^{-p-r} + (1 + \log n)n^{-p-1})$ uniformly. The sum of these differences is absolutely convergent, so grouping successive even and odd indices proves local uniform convergence of the original alternating series. Its individual terms are holomorphic. The same estimates apply to every termwise derivative, because derivatives increase p or r . This proves the assertion, including the conditionally convergent case $p = 1$. The stated absolute convergence follows directly from the bounds for $h_n^{(r)}$. $\square$

4.2 The exact bivariate identity

Theorem 4.2 (Two-shift reflection). *On*

$$\Omega = (\mathbb{C} \setminus \{\pm 3, \pm 5, \pm 7, \dots\}) \times (\mathbb{C} \setminus (\mathbb{Z} \setminus \{0\})),$$

one has

$$\boxed{A(z, u) + A(-z, -u) = B(z, u)}, \quad (4.4)$$

where

$$\begin{aligned} B(z, u) = & -\frac{\pi}{2 \cos(\pi z/2)} \left\{ \psi(1-u) - \psi\left(\frac{1+z}{2} - u\right) \right\} \\ & + \frac{\pi D(z-2u)}{\sin \pi u} - \frac{D(z)}{u}. \end{aligned} \quad (4.5)$$

Apparent singularities of the complete expression on Ω are removable. This convention applies in particular at $u = 0$, at $z = \pm 1$, and where an individual digamma term in (4.5) has a pole. There is no assertion that its summands can be evaluated separately at such points.

For $|z| < 3$ and $|u| < 1$, the ordinary Taylor expansion of A is

$$A(z, u) = \sum_{p,r \geq 1} Q_{p,r} z^{p-1} u^{r-1}. \quad (4.6)$$

Thus B determines precisely the terms with $p+r$ even in this germ.

Proof. We first prove the identity near $(z, u) = (0, 0)$. Set

$$G_r(z) = \int_0^1 \frac{x^{-z} \operatorname{Li}_r(-x^2)}{1+x^2} dx.$$

The harmonic generating function $\operatorname{Li}_r(w)/(1-w) = \sum_{n \geq 1} H_n^{(r)} w^n$ gives $G_r(z) = \sum_{p \geq 1} Q_{p,r} z^{p-1}$. One can first insert a common Abel factor and then pass to its limit; the preceding lemma identifies that limit with the ordinary sum. Alternatively, the logarithmic moment representation is

$$Q_{p,r} = \frac{1}{(p-1)!} \int_0^1 \frac{(-\log x)^{p-1} \operatorname{Li}_r(-x^2)}{1+x^2} dx. \quad (4.7)$$

For $0 \leq y \leq 1$ and integer $r \geq 1$, $|\operatorname{Li}_r(-y)| \leq y$ by the alternating-series test. It follows from (4.7) that $|Q_{p,r}| \leq 3^{-p}$. Hence the double power series in (4.6) converges normally for the stated radii. Its coefficients also follow by the justified derivatives of (4.2) at the origin.

The full Mellin identity, after the substitution $y = x^2$, is

$$M_r(z) := \int_0^\infty \frac{x^{-z} \operatorname{Li}_r(-x^2)}{1+x^2} dx = -\frac{\pi}{2 \cos(\pi z/2)} \left\{ \zeta\left(r, \frac{1+z}{2}\right) - \zeta(r) \right\}. \quad (4.8)$$

It initially suffices to take $|\Re z| < 1$. At $r = 1$, the expression in braces denotes the finite difference $\psi(1) - \psi((1+z)/2)$.

For $L \in \mathbb{R}$, define the polynomial

$$P_r(L) = \frac{(2L)^r}{r!} + 2 \sum_{j=1}^{\lfloor r/2 \rfloor} \eta(2j) \frac{(2L)^{r-2j}}{(r-2j)!}, \quad \eta(s) = (1 - 2^{1-s})\zeta(s). \quad (4.9)$$

Then

$$\operatorname{Li}_r(-e^{2L}) + (-1)^r \operatorname{Li}_r(-e^{-2L}) = -P_r(L). \quad (4.10)$$

Here the signs and branch convention can be checked without invoking a complex inversion formula. The derivative of the left side is twice the corresponding expression for $r - 1$. At $r = 0$ that expression equals -1 . At $L = 0$ it is zero for odd r , and $-2\eta(r)$ for even positive r . These data give exactly (4.9).

Splitting (4.8) at 1 and substituting $x \mapsto 1/x$ in its second part gives

$$G_r(z) - (-1)^r G_r(-z) = M_r(z) + (-1)^r \int_0^1 \frac{x^z P_r(\log x)}{1 + x^2} dx. \quad (4.11)$$

Now sum with multiplier u^{r-1} over $r \geq 1$. In a sufficiently small neighborhood, for example $|z| + 2|u| < 1$, all the integrations and sums involved are dominated by integrable powers of x at their endpoints. The two elementary generating identities are

$$\sum_{r \geq 1} u^{r-1} \{\zeta(r, b) - \zeta(r)\} = \psi(1 - u) - \psi(b - u), \quad (4.12)$$

$$\sum_{r \geq 0} P_r(L) u^r = \frac{\pi u}{\sin \pi u} e^{2uL}, \quad P_0 = 1. \quad (4.13)$$

In (4.12), the $r = 1$ value is interpreted as above; the formula follows by summing $(n + b - u)^{-1} - (n + 1 - u)^{-1}$ over $n \geq 0$. Equation (4.13) follows from $\pi u / \sin \pi u = 1 + 2 \sum_{j \geq 1} \eta(2j) u^{2j}$.

The sum of the left sides of (4.11) is $A(z, u) + A(-z, -u)$. Equation (4.12) gives the first line of (4.5); equation (4.13), with u replaced by $-u$, gives

$$\int_0^1 \frac{x^z}{1 + x^2} \left\{ \frac{\pi x^{-2u}}{\sin \pi u} - \frac{1}{u} \right\} dx = \frac{\pi D(z - 2u)}{\sin \pi u} - \frac{D(z)}{u}.$$

This proves the claimed identity as an analytic germ.

The paired series in Lemma 4.1 gives a meromorphic continuation of A to $\mathbb{C}^2$: near any candidate polar hyperplane, multiply by its local linear factors; the same uniform tail estimates apply. Its possible polar hyperplanes are $z = 3, 5, \dots$ and $u = 1, 2, \dots$. The right side of (4.5) is also meromorphic on $\mathbb{C}^2$. The identity theorem for meromorphic functions extends the equality from the germ to the whole connected domain. The left side is holomorphic on Ω , so all apparent poles of the complete right side there cancel. Finally, the sign in the Taylor projection is $(-1)^{(p-1)+(r-1)} = (-1)^{p+r}$, proving the final assertion. $\square$

Corollary 4.3 (Every reflected shifted harmonic sum). *For positive integers p, r and $(z, u) \in \Omega$,*

$$\begin{aligned} \sum_{n=1}^{\infty} (-1)^n \left\{ \frac{h_n^{(r)}(u)}{(2n+1-z)^p} + (-1)^{p+r} \frac{h_n^{(r)}(-u)}{(2n+1+z)^p} \right\} \\ = \frac{\partial_z^{p-1} \partial_u^{r-1} B(z, u)}{(p-1)!(r-1)!}. \end{aligned} \quad (4.14)$$

The right side is a finite expression in polygamma functions and elementary trigonometric functions. The equality holds for ordinary conditionally convergent sums when $p = 1$.

Proof. Differentiate (4.4). Lemma 4.1 justifies the operation, including $p = 1$. The identities $\partial_u^{r-1} h_n^{(1)}(u) = (r-1)! h_n^{(r)}(u)$ and $\partial_z^{p-1} (2n+1-z)^{-1} = (p-1)!(2n+1-z)^{-p}$ give the first term. Applying the chain rule to $A(-z, -u)$ contributes the factor $(-1)^{p+r}$. $\square$

At rational parameter pairs in the domain, the right side uses only polygamma values at rational arguments and elementary constants. The digamma differences reduce by Gauss's formula to logarithms of algebraic numbers and algebraic multiples of π . Positive polygamma orders reduce to Hurwitz zeta values, and hence by residue-class filters to depth-one polylogarithms at roots of unity. This gives an explicit depth-one expression for the reflected combination; it does not assert the same for either summand separately.

4.3 A finite formula for the unshifted parity coefficients

Write $\beta(s) = \sum_{n \geq 0} (-1)^n / (2n+1)^s$ for the Dirichlet beta function, initially on $\Re s > 0$ and continued analytically. For $j \geq 0$, define

$$d_0(r) = \begin{cases} 2 \log 2, & r = 1, \\ (2^r - 2) \zeta(r), & r > 1, \end{cases} \quad d_j(r) = \frac{(r)_j (2^{r+j} - 1) \zeta(r+j)}{2^j j!} \quad (j \geq 1),$$

and let $c_{2\ell} = |E_{2\ell}| \pi^{2\ell} / (2^{2\ell} (2\ell)!)$, where $\sec t = \sum_{\ell \geq 0} |E_{2\ell}| t^{2\ell} / (2\ell)!$. Put

$$C_{p,r} = 2^r \binom{p+r-1}{r} \beta(p+r) + 2 \sum_{j=1}^{\lfloor r/2 \rfloor} 2^{r-2j} \eta(2j) \binom{p+r-2j-1}{r-2j} \beta(p+r-2j). \quad (4.15)$$

The coefficient form of the reflection is

$$Q_{p,r} = \frac{(-1)^{r+1}}{2} \left\{ C_{p,r} - \frac{\pi}{2} \sum_{\ell=0}^{\lfloor (p-1)/2 \rfloor} c_{2\ell} d_{p-1-2\ell}(r) \right\}, \quad p+r \text{ even.} \quad (4.16)$$

Indeed, extract z^{p-1} in (4.11); the coefficient on its left is $2Q_{p,r}$ in the indicated parity. Taylor expansion of the Hurwitz zeta gives the d_j terms. Integration of each monomial in (4.9) gives the beta terms in (4.15); the common sign is $(-1)^{r+1}$. This is a finite coefficient algorithm for every p, r , with no integer-relation search.

For example, writing $G = \beta(2)$,

$$Q_{2,2} = \frac{7\pi}{4} \zeta(3) - 6\beta(4) - \frac{\pi^2 G}{12}, \quad (4.17)$$

$$Q_{1,3} = 4\beta(4) + \frac{\pi^2 G}{6} - \frac{3\pi}{2} \zeta(3), \quad (4.18)$$

$$Q_{4,2} = \frac{31\pi}{8} \zeta(5) + \frac{7\pi^3}{32} \zeta(3) - 20\beta(6) - \frac{\pi^2}{12} \beta(4), \quad (4.19)$$

$$Q_{3,3} = 40\beta(6) + \frac{\pi^2}{2} \beta(4) - \frac{93\pi}{8} \zeta(5) - \frac{3\pi^3}{16} \zeta(3), \quad (4.20)$$

$$Q_{4,4} = -280\beta(8) - \frac{10\pi^2}{3} \beta(6) - \frac{7\pi^4}{720} \beta(4) + \frac{635\pi}{8} \zeta(7) + \frac{31\pi^3}{16} \zeta(5). \quad (4.21)$$

Every term has total weight $p+r$. Only odd zeta values, even beta values, π , and, in the $r=1$ case, $\log 2$ occur.

4.4 A rational-shift identity with an elementary answer

Corollary 4.4. *The ordinary convergent nested sum satisfies*

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{2n+1} \sum_{j=1}^n \left\{ \frac{1}{j - \frac{1}{4}} + \frac{1}{j + \frac{1}{4}} \right\} = \pi(\log(1 + \sqrt{2}) - 1). \quad (4.22)$$

Equivalently, dividing by 32, the inner summand is $j/(16j^2 - 1)$ and the right side is $\pi(\log(1 + \sqrt{2}) - 1)/32$.

Proof. Set $(z, u) = (0, 1/4)$ in (4.4). The digamma reflection formula gives $\psi(3/4) - \psi(1/4) = \pi$, and $D(0) = \pi/4$. The remaining resolvent is

$$D(-1/2) = 2 \int_0^1 \frac{dt}{1+t^4} = \frac{\pi}{2\sqrt{2}} + \frac{\log(1+\sqrt{2})}{\sqrt{2}}.$$

The last integral follows by factoring $1+t^4 = (t^2 + \sqrt{2}t + 1)(t^2 - \sqrt{2}t + 1)$ and integrating the two partial fractions. Substitution into (4.5) gives $-\pi^2/2 + \pi\sqrt{2}D(-1/2) - \pi$, which is the stated answer. $\square$

4.5 Verification and the unresolved odd-weight direction

The accompanying `derive_parity.py` generates the finite symbolic formula (4.16). It independently compares all 25 ordered pairs with $p+r \leq 10$ and $p+r$ even against real quadrature of (4.7) at 60 decimal working digits. The largest absolute discrepancy is less than 3×10^{-58} . Three separate 450-term Euler evaluations of (4.2), including a complex parameter pair and a point outside the initial small-parameter proof neighborhood, agree with (4.5) to more than 115 decimal places. The JSON file records every tested pair and residual. Three mixed-derivative checks, including $p=1, r=2$, and the quarter-shift specialization are checked by the same independent series method; their absolute discrepancies are below 3×10^{-136} . These checks audit the signs and normalizations; the arbitrary-order proof is the analytic argument above.

At $(z, u) = (0, 0)$, equation (4.14) has coefficient $1 + (-1)^{p+r}$. It therefore gives no value for $Q_{p,r}$ when $p+r$ is odd. In particular, $S_6 = Q_{6,1}$ and $S_8 = Q_{8,1}$ remain outside the scalar parity projection. This is a precise limitation of the reflection method, not a proof of arithmetic irreducibility. The repository already exhibits a nonzero separating functional for the restricted depth-two product presentation of the S_6 candidate; the present identity does not supply the missing odd-weight relation.

5 Dilation and unequal-frequency correlations

Two explicit questions in the incoming reports concern the effect of integer dilation on canonical periodic finite parts, and products with unequal integer dilations. They occur in the further-research sections of the shifted-Hurwitz and convolution-calculus deliveries. This section answers those questions for every nonnegative Stieltjes index and every nonnegative parameter derivative order, provided the two sets of singular points are disjoint. The location of the surviving ordinary special-function values is determined by the reduced dilation ratio; their coefficients also remember the physical coordinate used for the finite part.

5.1 Conventions and the elementary contact identity

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and use normalized Lebesgue measure. The spectral convention is

$$\zeta(1-z, x) = -\frac{1}{z} + \sum_{n \geq 0} \gamma_n(x) \frac{z^n}{n!}, \quad \gamma_0(x) = -\psi(x), \quad 0 < x < 1.$$

For a smooth periodic test function ϕ , let $T_{n,r}$ be the constant-term extension of $\gamma_n^{(r)}(x)$ from $(0, 1)$:

$$\langle T_{n,r}, \phi \rangle = \text{CT}_{\epsilon \downarrow 0} \int_{\epsilon}^1 \gamma_n^{(r)}(x) \phi(x) dx.$$

Here CT retains the constant term after all negative powers of ϵ multiplied by powers of $\log \epsilon$, and all positive powers of $\log \epsilon$, have been removed. No finite point-supported term is added.

Thus, with $G_n = T_{n,0}$,

$$\langle G_n, \phi \rangle = \int_0^1 \left\{ \gamma_n(x) \phi(x) - \frac{\log^n x}{x} \phi(0) \right\} dx.$$

The shift identity $\gamma_n(x) = x^{-1} \log^n x + \gamma_n(1+x)$ proves convergence and the equality of these definitions.

Put

$$p_r(z) = \prod_{j=1}^r \left(1 - \frac{z}{j} \right), \quad h_{r,n} = (-1)^n n! e_{n+1}(1, 1/2, \dots, 1/r), \quad (5.1)$$

where the empty product is one and elementary symmetric functions of degree greater than r are zero. The undilated contact identity, proved in the incoming convolution and shifted-Hurwitz reports [10, 9], is

$$D^r G_n = T_{n,r} - h_{r,n} \delta_0^{(r)}. \quad (5.2)$$

For completeness, its origin can be read directly from the meromorphic periodic Hurwitz family. If

$$\widehat{R}_z(0) = 0, \quad \widehat{R}_z(k) = (2\pi i k)^{-z} \quad (k \neq 0), \quad Z(1-z) = \Gamma(z) R_z,$$

then

$$Z(1-z) = \frac{\delta_0 - 1}{z} + \sum_{n \geq 0} G_n \frac{z^n}{n!}.$$

For $r \geq 1$, the local singular part of the pointwise r th derivative is $(-1)^r r! p_r(z) x^{z-r-1}$. The meromorphic extension of x^{z-r-1} has residue $(-1)^r \delta_0^{(r)} / r!$ at $z = 0$. Consequently its polar contribution is $p_r(z) \delta_0^{(r)} / z$. Comparing coefficients with $D^r Z(1-z)$ gives (5.2), since $n! [z^{n+1}] p_r(z) = -h_{r,n}$. The case $r = 0$ is immediate. This derivation also specifies all signs. The coefficients can equally be written through generalized harmonic numbers: for $|z| < 1$,

$$p_r(z) = \exp \left(- \sum_{j \geq 1} \frac{H_r^{(j)}}{j} z^j \right), \quad H_r^{(j)} = \sum_{\ell=1}^r \ell^{-j}.$$

For example $e_1 = H_r$ and $e_2 = (H_r^2 - H_r^{(2)})/2$.

5.2 The exact dilation anomaly

For a positive integer q , let U_q be pullback by the covering map $x \mapsto qx$:

$$\langle U_q T, \phi \rangle = \left\langle T, \frac{1}{q} \sum_{j=0}^{q-1} \phi \left(\frac{x+j}{q} \right) \right\rangle.$$

For an ordinary function this is $U_q f(x) = f(qx)$. Write

$$\Delta_q = \sum_{j=0}^{q-1} \delta_{j/q}, \quad U_q \delta_0^{(r)} = q^{-r-1} \Delta_q^{(r)}.$$

Let $X_{q,n,r}$ be the constant-term extension of $\gamma_n^{(r)}(\{qx\})$ using the *ambient coordinate* x : at each grid point j/q , delete the right interval $(j/q, j/q + \epsilon)$ and take the constant term as ϵ tends to zero. The transported finite part $U_q T_{n,r}$ takes the constant term in a source cutoff η , corresponding to ambient cutoff η/q . The ambient finite part instead takes the constant term in ϵ after the source lower limit becomes $q\epsilon$. The variable in which the constant term is extracted is part of the normalization.

Theorem 5.1 (All-order finite-part dilation law). *For $q \geq 1$ and $n, r \geq 0$, define the explicit polynomial*

$$c_{n,r}(L) = n![z^{n+1}]p_r(z)(e^{Lz} - 1) = n! \sum_{j=0}^{\min(r,n)} \frac{(-1)^j e_j(1, 1/2, \dots, 1/r)}{(n+1-j)!} L^{n+1-j}. \quad (5.3)$$

Then

$$\boxed{X_{q,n,r} = U_q T_{n,r} - q^{-r-1} c_{n,r}(\log q) \Delta_q^{(r)}}. \quad (5.4)$$

In particular $c_{n,0}(L) = L^{n+1}/(n+1)$ and $c_{0,r}(L) = L$.

Proof. Near a grid point use $t = x - j/q > 0$. The singular part of the generating function for the argument derivatives is

$$(-1)^r r! p_r(z) (qt)^{z-r-1} = (-1)^r r! q^{-r-1} p_r(z) e^{z \log q} t^{z-r-1}.$$

After pairing with a test function, only its Taylor term of degree r produces a pole at $z = 0$. Its contribution is

$$q^{-r-1} \frac{p_r(z) e^{z \log q}}{z} \delta_{j/q}^{(r)}.$$

The pullback of the source-coordinate family instead has polar contribution $q^{-r-1} p_r(z) \delta_{j/q}^{(r)}/z$. The regular Taylor coefficients left after subtracting these complete polar contributions are respectively the ambient-coordinate constant terms and the pulled-back constant terms. Their difference is therefore the negative of

$$q^{-r-1} p_r(z) \frac{e^{z \log q} - 1}{z} \delta_{j/q}^{(r)}.$$

Extract $z^n/n!$ and sum over the grid. When $r = 0$, the scalar spectral pole $-1/z$ occurs in both calculations and cancels. Subtracting the test-function Taylor polynomial through degree r makes every remaining local integral holomorphic near $z = 0$, which justifies the coefficient comparison. Expanding the finite product gives (5.3). $\square$

Corollary 5.2 (Differentiation and dilation with fixed ambient scale). *Define*

$$b_{r,n}(L) = \sum_{j=0}^n \binom{n}{j} L^{n-j} h_{r,j}. \quad (5.5)$$

Then

$$\boxed{D^r X_{q,n,0} = q^r X_{q,n,r} - \frac{b_{r,n}(\log q)}{q} \Delta_q^{(r)}}. \quad (5.6)$$

For every $n \geq 0$ this specializes to

$$DX_{q,n,0} = q X_{q,n,1} - \frac{(\log q)^n}{q} \Delta'_q, \quad (5.7)$$

where $(\log 1)^0 = 1$.

Proof. Differentiate (5.4) for $r = 0$, use $D^r U_q = q^r U_q D^r$ and (5.2), and compare with (5.4) at general r . The remaining scalar identity is

$$b_{r,n}(L) = h_{r,n} + \frac{L^{n+1}}{n+1} - c_{n,r}(L) = n![z^{n+1}](1 - p_r(z))e^{Lz}.$$

For $r = 1$, only $h_{1,0} = 1$ is nonzero. $\square$

The undilated anomaly vanishes when $n \geq r$. Equation (5.7) shows that this vanishing generally fails after ambient-coordinate dilation: already $n = r = 1$ gives $DX_{q,1,0} = qX_{q,1,1} - (\log q)\Delta'_q/q$. This is an extension of the incoming contact theorem, not a correction to its correctly normalized undilated statement.

The scale corrections are compatible with successive dilations. The following polynomial identities hold for all $L, M \in \mathbb{C}$:

$$\begin{aligned} c_{n,r}(L+M) &= c_{n,r}(M) + \sum_{j=0}^n \binom{n}{j} M^{n-j} c_{j,r}(L), \\ b_{r,n}(L+M) &= \sum_{j=0}^n \binom{n}{j} M^{n-j} b_{r,j}(L). \end{aligned} \quad (5.8)$$

Indeed, use $e^{(L+M)z} - 1 = (e^{Mz} - 1) + e^{Mz}(e^{Lz} - 1)$ in (5.3), and use the last generating expression in the preceding proof for b . Thus a two-step change of local scale gives the same finite correction as the combined scale; no independent finite constant is introduced by composing the operations.

5.3 A multiplication theorem including the point masses

Let $\tau_a T(x) = T(x+a)$ and $L = \log q$.

Theorem 5.3 (Distributional multiplication law). *For $n \geq 0$,*

$$\sum_{j=0}^{q-1} \tau_{j/q} G_n = q \sum_{i=0}^n \binom{n}{i} (-L)^{n-i} U_q G_i + \frac{(-L)^{n+1}}{n+1} (\Delta_q - q). \quad (5.9)$$

More generally, for $r \geq 1$,

$$\sum_{j=0}^{q-1} \tau_{j/q} T_{n,r} = q^{r+1} \sum_{i=0}^n \binom{n}{i} (-L)^{n-i} U_q T_{i,r} + c_{n,r}(-L) \Delta_q^{(r)}. \quad (5.10)$$

Proof. The classical Hurwitz multiplication formula gives, as meromorphic periodic distributions,

$$\sum_{j=0}^{q-1} \tau_{j/q} Z(1-z) = q^{1-z} U_q Z(1-z).$$

It can also be proved directly from the Fourier coefficients: the sum annihilates frequencies not divisible by q . Expanding the distributional Laurent series proves (5.9); the residue transforms as $\sum \tau_{j/q}(\delta_0 - 1) = \Delta_q - q$.

Differentiate the pointwise multiplication formula r times and then take ambient-coordinate constant terms. Substitution of (5.4) reduces the additional coefficient to

$$-\sum_{i=0}^n \binom{n}{i} (-L)^{n-i} c_{i,r}(L) = n! [z^{n+1}] p_r(z) (e^{-Lz} - 1) = c_{n,r}(-L).$$

This gives (5.10). □

5.4 The undilated kernel and an explicit finite coefficient rule

The unequal-dilation theorem below uses a known undilated closure from the incoming reports [11, 9]. We record its generating function and derivation so that the resulting algorithm is self-contained. These undilated statements are prior results within the supplied material.

For $0 < t < 1$ and $\Re s, \Re v < 1$, set

$$C(s, v; t) = \int_0^1 \zeta(s, x) \zeta(v, \{x + t\}) dx.$$

The integral is ordinary: the two singularities are separated. Its two equivalent continuations are

$$C(s, v; t) = \Gamma(1-s)\Gamma(1-v)(2\pi)^{s+v-2} \{e^{-\pi i(s-v)/2} \text{Li}_{2-s-v}(e^{2\pi i t}) + e^{\pi i(s-v)/2} \text{Li}_{2-s-v}(e^{-2\pi i t})\}, \quad (5.11)$$

$$C(s, v; t) = \Gamma(s+v-1) \left\{ \frac{\Gamma(1-s)}{\Gamma(v)} \zeta(s+v-1, t) + \frac{\Gamma(1-v)}{\Gamma(s)} \zeta(s+v-1, 1-t) \right\}. \quad (5.12)$$

All apparent singularities are interpreted in the complete expression. To prove the first formula, start with $\Re s, \Re v < 0$ and the absolutely convergent Hurwitz Fourier series, whose nonzero coefficients are

$$\hat{\zeta}_s(k) = \Gamma(1-s)(2\pi i k)^{s-1}, \quad k \neq 0.$$

The zero coefficient is zero. Integrating the product retains the opposite frequencies and yields (5.11). For the second formula, insert the same Fourier expansion with spectral order $s+v-1$ on its right. Euler's gamma reflection identity reduces the coefficient of the positive-argument polylogarithm by

$$\frac{\sin(\pi s)e^{-\pi i(s+v)/2} + \sin(\pi v)e^{\pi i(s+v)/2}}{\sin \pi(s+v)} = e^{-\pi i(s-v)/2}.$$

The other coefficient is obtained by reversing the signs in the phases. This calculation holds off the integer hyperplanes; analytic continuation extends it across their removable singularities. The local decomposition $\zeta(s, x) = x^{-s} + \zeta(s, 1+x)$ justifies joint holomorphy of the integral on $\Re s, \Re v < 1$, and hence the continuation to that region. This is the classical Fourier and product-integral mechanism [4, 7] in the circular-translation convention.

Define, near $(u, v) = (0, 0)$,

$$\begin{aligned} \mathcal{A}(u, v) &= \frac{\Gamma(1+u+v)\Gamma(1-v)}{\Gamma(1+u)}, & \mathcal{B}(u, v) &= \mathcal{A}(v, u), \\ \mathcal{K}(u, v; t) &= uv C(1+u, 1+v; t). \end{aligned}$$

Equation (5.12) gives

$$\mathcal{K}(u, v; t) = -u\mathcal{A}(u, v)\zeta(1+u+v, 1-t) - v\mathcal{B}(u, v)\zeta(1+u+v, t). \quad (5.13)$$

This function is holomorphic at the origin. Indeed, $\mathcal{A}(u, -u) = \mathcal{B}(u, -u) = 1$, so $(u\mathcal{A} + v\mathcal{B})/(u+v)$ has a removable denominator. Removing the common Hurwitz pole in (5.13) leaves the negative of that quotient plus analytic Stieltjes series.

For the canonical undilated finite part one consequently has the terminating coefficient rule

$$\boxed{I_{mn}(t) = (-1)^{m+n} m!n! [u^{m+1}v^{n+1}] \mathcal{K}(u, v; t).} \quad (5.14)$$

Here I_{mn} denotes the constant term of $\int \gamma_m(x) \gamma_n(\{x+t\}) dx$ with the two ambient cutoffs. For clarity, the matching to a finite part can be justified before extracting any coefficients. Put $\mathcal{H}(u, x) = \zeta(1+u, x) - 1/u$ and $b = 1-t$. The following locally convergent expression is

analytic for small u, v :

$$\begin{aligned} & \int_0^b \left\{ x^{-1-u} [\mathcal{H}(v, t+x) - \mathcal{H}(v, t)] + \mathcal{H}(u, 1+x) \mathcal{H}(v, t+x) \right\} dx \\ & + \int_0^t \left\{ y^{-1-v} [\mathcal{H}(u, b+y) - \mathcal{H}(u, b)] + \mathcal{H}(v, 1+y) \mathcal{H}(u, b+y) \right\} dy \\ & + \mathcal{H}(v, t) \frac{1-b^{-u}}{u} + \mathcal{H}(u, b) \frac{1-t^{-v}}{v}. \end{aligned}$$

The bracketed differences vanish linearly; both quotients have removable values at zero. The coefficient of $u^m v^n$ is exactly $(-1)^{m+n} I_{mn}(t)/(m!n!)$, by direct integration of the subtracted logarithmic singularities. In a left spectral half-plane the displayed expression is also

$$C(1+u, 1+v; t) + \frac{1}{uv} + \frac{\mathcal{H}(v, t)}{u} + \frac{\mathcal{H}(u, b)}{v}.$$

Using the axis values of (5.13), it equals

$$\frac{\mathcal{K}(u, v; t) - \mathcal{K}(u, 0; t) - \mathcal{K}(0, v; t) + \mathcal{K}(0, 0; t)}{uv}.$$

This proves (5.14) with the specified normalization.

The coefficient calculation uses rational polynomial arithmetic and positive integer zeta values only, because

$$\log \mathcal{A}(u, v) = \sum_{j \geq 2} \frac{\zeta(j)}{j} \left\{ (-1)^j ((u+v)^j - u^j) + v^j \right\}. \quad (5.15)$$

For an explicit finite algorithm, let ℓ_j be the homogeneous summand of degree j here. The homogeneous terms of $\mathcal{A}$ satisfy

$$\mathcal{A}_0 = 1, \quad \mathcal{A}_1 = 0, \quad \mathcal{A}_d = \frac{1}{d} \sum_{j=2}^d j \ell_j \mathcal{A}_{d-j}.$$

Only $d \leq m+n+2$ can contribute to (5.14). This proves finite linear closure in the single values $\gamma_j(t)$ and $\gamma_j(1-t)$, with $j \leq m+n+1$. Their top coefficients are $1/(m+1)$ and $1/(n+1)$; the coefficients at order $m+n$ vanish since $\mathcal{A}_1 = 0$. These are the incoming closure theorem and its missing-order rule, now available as an explicit input to the new dilation calculation.

5.5 Unequal integer dilations: the full Stieltjes theorem

Fix positive integers p, q , put

$$d = \gcd(p, q), \quad P = p/d, \quad Q = q/d, \quad \theta = \{Pa\}, \quad 0 < \theta < 1. \quad (5.16)$$

The last condition is exactly the condition that the singular grids of $\gamma_m(\{px\})$ and $\gamma_n(\{qx+a\})$ do not intersect. Indeed an intersection would give $qx+a \in \mathbb{Z}$ and $px \in \mathbb{Z}$, equivalently $Pa \in \mathbb{Z}$ since $(P, Q) = 1$. Let

$$F_{mn}^{p,q}(\theta) = \text{FP}_x \int_0^1 \gamma_m(\{px\}) \gamma_n(\{qx+a\}) dx.$$

The notation FP_x means the common ambient-coordinate constant term specified above. One may equivalently take a separate constant term in an independent ambient cutoff variable at

each grid point. Replacing those variables by $c_j\epsilon$ and taking a single constant term would generally change the answer through local $\log c_j$ terms.

We use the already established undilated correlation

$$I_{mn}(t) = \text{FP} \int_0^1 \gamma_m(x) \gamma_n(\{x+t\}) dx, \quad 0 < t < 1.$$

Its all-order single-Stieltjes closure and finite coefficient rule were recalled in Section 5.4. The formulas needed below are

$$I_{00}(t) = \gamma_1(t) + \gamma_1(1-t) - 2\zeta(2), \quad (5.17)$$

$$I_{10}(t) = \frac{1}{2}\gamma_2(t) + \gamma_2(1-t) + \zeta(2)\{\gamma_0(t) + \gamma_0(1-t)\} - \zeta(3). \quad (5.18)$$

For $M \geq 1$ define a normalized trace polynomial

$$A_n(M, t) = \sum_{j=0}^n \binom{n}{j} (-\log M)^{n-j} \gamma_j(t) - \frac{(-\log M)^{n+1}}{n+1}. \quad (5.19)$$

In particular $A_0(M, t) = \gamma_0(t) + \log M$. The pointwise Hurwitz multiplication identity [4, §25.11] proves

$$\frac{1}{M} \sum_{j=0}^{M-1} \gamma_n\left(\left\{u + \frac{j}{M}\right\}\right) = A_n(M, \{Mu\}), \quad Mu \notin \mathbb{Z}.$$

Theorem 5.4 (Unequal-dilation Stieltjes closure). *Set $A = \log P$, $B = \log Q$, $L_p = \log p$, and $L_q = \log q$. For all $m, n \geq 0$ and the parameters (5.16),*

$$\begin{aligned} F_{mn}^{p,q}(\theta) &= \sum_{i=0}^m \sum_{j=0}^n \binom{m}{i} \binom{n}{j} (-B)^{m-i} (-A)^{n-j} I_{ij}(\theta) \\ &\quad + \frac{(-B)^{m+1} - L_p^{m+1}}{m+1} A_n(P, \theta) + \frac{(-A)^{n+1} - L_q^{n+1}}{n+1} A_m(Q, 1-\theta) \\ &\quad + \frac{(-B)^{m+1} (-A)^{n+1}}{(m+1)(n+1)}. \end{aligned} \quad (5.20)$$

Thus every such product is a finite linear combination of single Stieltjes values at θ and $1-\theta$, with coefficients in the algebra generated over $\mathbb{Q}$ by the logarithms of p, q, P, Q and positive integer zeta values.

Proof. Let $C(s, t; \theta)$ be the ordinary translated Hurwitz correlation, initially for $\Re s, \Re t < 1$. Fourier orthogonality first in $\Re s, \Re t < 0$ gives

$$\int_0^1 \zeta(s, \{px\}) \zeta(t, \{qx+a\}) dx = Q^{s-1} P^{t-1} C(s, t; \theta). \quad (5.21)$$

Indeed the surviving frequency indices satisfy $pu + qv = 0$, hence $u = Q\ell$, $v = -P\ell$; their translation factor is $e^{-2\pi i \ell P a}$. The two sides have the same meromorphic continuation after local endpoint subtraction. The common divisor d does not occur in (5.21), because ordinary integration is unchanged by a common covering.

Put $\mathcal{G}(z, x) = \sum_{n \geq 0} \gamma_n(x) z^n / n!$ and $\mathcal{I}(z, w; \theta) = \sum_{m, n \geq 0} I_{mn}(\theta) z^m w^n / (m!n!)$. For distinct singular points the distributional Laurent series gives

$$C(1-z, 1-w; \theta) = -\frac{1}{zw} + \frac{\mathcal{G}(w, \theta)}{z} + \frac{\mathcal{G}(z, 1-\theta)}{w} + \mathcal{I}(z, w; \theta). \quad (5.22)$$

For example, the coefficient $-1/(zw)$ is the correlation of $\delta_0 - 1$ with its nontrivial translate. Omitting it would change the final constant in (5.20).

Multiply (5.22) by e^{-Bz-Aw} and extract $z^m w^n / (m!n!)$. This computes the correlation of the *true pullbacks* $U_p G_m$ and $G_n(qx + a)$. Its value is the double sum in (5.20), plus

$$\frac{(-B)^{m+1}}{m+1} A_n(P, \theta) + \frac{(-A)^{n+1}}{n+1} A_m(Q, 1 - \theta) + \frac{(-B)^{m+1}(-A)^{n+1}}{(m+1)(n+1)}.$$

The $r = 0$ dilation law changes the first factor by $-L_p^{m+1} \Delta_p / (p(m+1))$. Pairing this with the other factor gives $-L_p^{m+1} A_n(P, \theta) / (m+1)$: the grid values form d copies of a complete reduced P -grid. The second factor contributes $-L_q^{n+1} A_m(Q, 1 - \theta) / (n+1)$ similarly. The product of the two grid distributions vanishes, because their supports are disjoint. Their sum is (5.20). The product and all coefficient extractions can be justified using a partition of unity separating the two singular grids: on each singular neighborhood the other factor is smooth, and elsewhere both factors are smooth. $\square$

For example, if $p = 2$, $q = 1$, $\theta = \{2a\} \in (0, 1)$ and $L = \log 2$, (5.20) gives the explicit new family member

$$\begin{aligned} \text{FP}_x \int_0^1 \gamma_1(\{2x\}) \gamma_0(\{x+a\}) dx &= \frac{1}{2} \gamma_2(\theta) + \gamma_2(1 - \theta) - L \gamma_1(1 - \theta) \\ &\quad + \left(\zeta(2) - \frac{1}{2} L^2 \right) \gamma_0(\theta) + \zeta(2) \gamma_0(1 - \theta) - \zeta(3) - \frac{1}{2} L^3. \end{aligned} \quad (5.23)$$

5.6 Every mixed parameter derivative

Theorem 5.5 (Dilated differentiation and product integration). *Let $N = r + k$ with $r, k, m, n \geq 0$, and differentiate $F_{mn}^{p,q}$ in its displayed argument θ . Then*

$$\begin{aligned} &\text{FP}_x \int_0^1 \gamma_m^{(r)}(\{px\}) \gamma_n^{(k)}(\{qx+a\}) dx \\ &= P^k Q^r \left\{ (-1)^r \frac{d^N}{d\theta^N} F_{mn}^{p,q}(\theta) + (-1)^r b_{r,m}(L_p) \partial_\theta^N A_n(P, \theta) \right. \\ &\quad \left. + (-1)^k b_{k,n}(L_q) \partial_t^N A_m(Q, t) \Big|_{t=1-\theta} \right\}. \end{aligned} \quad (5.24)$$

The derivatives on the left are derivatives in the special-function argument, not derivatives of the fractional-part map.

Proof. Rewrite (5.6) as

$$X_{p,m,r} = p^{-r} D^r X_{p,m,0} + p^{-r-1} b_{r,m}(L_p) \Delta_p^{(r)},$$

and use the translated counterpart for the second factor. Periodic integration by parts in the first product yields

$$p^{-r} q^{-k} \int (D^r X_{p,m,0})(D^k X_{q,n,0}(\cdot + a/q)) = (-1)^r (q/p)^r \partial_a^N F_{mn}^{p,q}(\{Pa\}).$$

Since $\partial_a = P \partial_\theta$ on each noncollision interval, this is $(-1)^r P^k Q^r \partial_\theta^N F$. The first grid contact pairs with the smooth second factor and gives

$$(-1)^r (q/p)^r b_{r,m}(L_p) \frac{1}{p} \sum_{j=0}^{p-1} \gamma_n^{(N)}(\{qj/p + a\}) = (-1)^r P^k Q^r b_{r,m}(L_p) \partial_\theta^N A_n(P, \theta).$$

The second contact is the third term in (5.24). The grid-grid term vanishes by disjointness. The same separated partition of unity used in the preceding proof makes these integrations by parts valid. $\square$

5.7 A particularly short polygamma identity

Put $H_0 = 0$, $H_j = \sum_{\ell=1}^j 1/\ell$ for $j \geq 1$, and

$$K = \log \text{lcm}(p, q).$$

Corollary 5.6 (All unequal-dilation polygamma correlations). *For $N = r + k \geq 1$,*

$$\begin{aligned} & \text{FP}_x \int_0^1 \psi^{(r)}(\{px\}) \psi^{(k)}(\{qx + a\}) dx \\ &= N! P^k Q^r \left\{ (-1)^{k+1} [\zeta'(N+1, \theta) + (H_N - H_r + K) \zeta(N+1, \theta)] \right. \\ & \quad \left. + (-1)^{r+1} [\zeta'(N+1, 1-\theta) + (H_N - H_k + K) \zeta(N+1, 1-\theta)] \right\}. \end{aligned} \quad (5.25)$$

The two primes on the right denote first derivatives in the spectral variable of Hurwitz zeta. The $r = k = 0$ case is

$$\begin{aligned} & \text{FP}_x \int_0^1 \psi(\{px\}) \psi(\{qx + a\}) dx \\ &= \gamma_1(\theta) + \gamma_1(1-\theta) + K \{\psi(\theta) + \psi(1-\theta)\} - \frac{\pi^2}{3} - K^2 + \log p \log q. \end{aligned} \quad (5.26)$$

Proof. Apply (5.20) with $m = n = 0$. Its logarithmic coefficient is $K = L_p + B = L_q + A$, and its constant simplifies using $AB - K(A + B) = -K^2 + L_p L_q$. This proves (5.26). In (5.24), use $b_{r,0} = H_r$, $\gamma_0 = -\psi$, and the pointwise identities

$$\psi^{(N)}(t) = (-1)^{N+1} N! \zeta(N+1, t), \quad \gamma_1^{(N)}(t) = (-1)^{N+1} N! \{\zeta'(N+1, t) + H_N \zeta(N+1, t)\}.$$

The two reflection signs are $(-1)^r$ and $(-1)^k$ before this conversion. Simplification gives (5.25). $\square$

The least common multiple in the logarithm is forced by the local normalization. For example, a common dilation $p = q = d$ leaves the ordinary spectral correlation unchanged, but its digamma finite part changes by $\log d \{\psi(\theta) + \psi(1-\theta)\}$. Discarding this term would confuse transported finite parts with ambient-coordinate finite parts.

Corollary 5.7 (Odd total derivative order at a half shift). *If $\{Pa\} = 1/2$ and $N = r + k$ is odd, then*

$$\begin{aligned} & \text{FP}_x \int_0^1 \psi^{(r)}(\{px\}) \psi^{(k)}(\{qx + a\}) dx \\ &= N! P^k Q^r (-1)^{k+1} (H_k - H_r) (2^{N+1} - 1) \zeta(N+1). \end{aligned} \quad (5.27)$$

In particular every such value is a rational multiple of π^{N+1} . One unequal-dilation example is

$$\text{FP}_x \int_0^1 \psi(\{2x\}) \psi'(\{3x + 1/4\}) dx = \pi^2. \quad (5.28)$$

Proof. When N is odd the two signs in (5.25) are opposite. At $\theta = 1/2$ the spectral derivatives, the H_N terms, and the K terms cancel pairwise. The surviving harmonic difference multiplies $\zeta(N+1, 1/2) = (2^{N+1} - 1) \zeta(N+1)$. Since $N+1$ is even, Euler's even-zeta formula gives the last assertion. $\square$

The same unequal pair at zero derivative order gives a different exact identity, now involving the ordinary first Stieltjes constant:

$$\begin{aligned} & \text{FP}_x \int_0^1 \psi(\{2x\})\psi(\{3x + 1/4\}) dx \\ &= 2\gamma_1 - 2\gamma \log 24 - \frac{\pi^2}{3} - 7\log^2 2 - 5\log 2 \log 3 - \log^2 3. \end{aligned} \quad (5.29)$$

This follows from (5.26), $\psi(1/2) = -\gamma - 2\log 2$ and $\gamma_1(1/2) = \gamma_1 - 2\gamma \log 2 - \log^2 2$.

5.8 Ordinary convergent-integral realization

Every identity in Corollary 5.6 can be written without a finite-part symbol. Sort the two disjoint singular grids as $0 = x_0 < x_1 < \dots < x_{M-1} < 1$, put $x_M = 1$, and set $\ell_i = x_{i+1} - x_i$. On each interval use the ordinary branches

$$f_i(t) = \psi^{(r)}(\alpha_i + pt)\psi^{(k)}(\beta_i + qt), \quad 0 < t < \ell_i,$$

where $\alpha_i = \{px_i\}$ and $\beta_i = \{qx_i + a\}$, with the right-hand branch chosen at zero. Exactly one of α_i, β_i is zero. If $\alpha_i = 0$, set $v_i = r$ and

$$a_{ij} = \frac{(-1)^{r+1}r!q^j}{p^{r+1}j!}\psi^{(k+j)}(\beta_i), \quad 0 \leq j \leq r.$$

If $\beta_i = 0$, set $v_i = k$ and

$$a_{ij} = \frac{(-1)^{k+1}k!p^j}{q^{k+1}j!}\psi^{(r+j)}(\alpha_i), \quad 0 \leq j \leq k.$$

Then the left side of (5.25), including its $N = 0$ counterpart, equals the ordinary absolutely convergent expression

$$\sum_{i=0}^{M-1} \left[\int_0^{\ell_i} \left\{ f_i(t) - \sum_{j=0}^{v_i} a_{ij} t^{j-v_i-1} \right\} dt + a_{i,v_i} \log \ell_i + \sum_{j=0}^{v_i-1} \frac{a_{ij} \ell_i^{j-v_i}}{j - v_i} \right]. \quad (5.30)$$

This follows by integrating the displayed singular monomials explicitly. The subtracted integrands are bounded at their left endpoints. At the right endpoint the branch approaching one is used, so no singularity is introduced there.

The independent numerical implementation uses precisely (5.30) on the left and (5.25)–(5.26) on the right. Nine cases include unequal dilations, a nontrivial common divisor, and derivative orders up to $r + k = 3$. At 42 decimal working digits the largest relative discrepancy is below 3.5×10^{-39} . These are floating-point consistency checks of formulas proved above, not certified error enclosures or a substitute for the analytic argument. Three further independent subtracted-integral checks use positive Stieltjes indices $(m, n) = (1, 0), (0, 1), (1, 1)$, the last with $(p, q) = (2, 3)$. At 28 working digits their largest relative discrepancy is below 9×10^{-29} .

6 Research status, corrections, and further questions

6.1 What the continuation settles

The main resolved question in the supplied incoming set concerns integer dilation and the choice of local subtraction coordinate. Theorem 5.1 gives the complete local correction, including every Stieltjes index and argument derivative. Equations (5.20) and (5.24) then evaluate unequal-dilation products when the singular grids are disjoint. Their dependence on

$\log \text{lcm}(p, q)$ is explicit. These statements answer the integer-covering case of the rescaling question; a general nonlinear coordinate change is outside their scope.

The exact locations are the further-research items on unequal dilations in *Bilinear Closure of Generalized Stieltjes Correlations* [11], on dilation, traces and subtraction scale in *Shifted Hurwitz-Zeta Correlations* [9], and on rescaling in *A Convolution Calculus for Stieltjes Functions* [12]. The integration notes give the archive members and line locations in the pinned snapshot.

The Mellin results are independent of that regularization problem. They give ordinary integrals with a uniform finite formula at arbitrary denominator order. The integer resonances, all their logarithmic moments, and the spectral derivatives in Section 3 are proved identities. The harmonic reflection is likewise an identity of holomorphic functions and ordinary convergent series. Its even-weight unshifted coefficient evaluations have published antecedents [6]; the two-shift formulation is presented with a proof and without an unverified claim of historical priority.

Claim	Status after this work
All- N Mellin transform and integer resonances	Proved in Theorems 2.3 and 2.4.
Shifted generalized-harmonic reflection	Proved in Theorem 4.2; scalar parity precedent credited.
Integer-dilation finite parts and disjoint-grid products	Proved in Theorems 5.1, 5.4, and 5.5.
Undilated Stieltjes closure and convolution	Already proved in the supplied incoming material; recalled as inputs.
Gaussian S_4 evaluation	Already proved in the pinned manuscript; its status is unchanged.
Current Gaussian S_6 and S_8 evaluations	Still conjectural. No equality follows from a small residual.
Colliding unequal singular grids; nonlinear coordinate changes	Not treated by the new dilation theorem.

6.2 The Gaussian conjectures and the missing parity component

The manuscript defines

$$S_p = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^p}, \quad g_{a,b} = \text{Im Li}_{a,b}(i, 1),$$

where $\text{Li}_{a,b}(x, y) = \sum_{n > m \geq 1} x^n y^m / (n^a m^b)$ in its convergence region. Its proved translation is

$$S_p = \text{Im}(\text{Li}_{p,1}(i, 1) + \text{Li}_{p,1}(i, -1)).$$

The file `04-S4-proof.tex` supplies the exact S_4 certificate and a convergent octahedral relation. The current S_6 conjecture is in `04-cyclotomic-quotients.tex`; the current S_8 conjecture is in `04-S8-candidate.tex`. Their complete coefficient vectors and source labels are preserved in `integration/GAUSSIAN_STATUS.md`.

For specificity, the S_6 equality still to be proved is

$$\begin{aligned} S_6 \stackrel{?}{=} & \frac{722}{527} g_{6,1} + \frac{40}{527} g_{4,3} - \frac{128}{155} g_{2,5} + \frac{15191}{28569600} \pi^7 \\ & - \frac{3}{124} \beta(2) \zeta(5) - \frac{2373}{10540} \beta(4) \zeta(3) - 2\beta(6) \log 2. \end{aligned} \quad (6.1)$$

The manuscript also proves a change of coordinates between two presentations of this candidate. That identity equates two residual expressions; it does not show that either residual is zero.

The obstruction to proving (6.1) by the reflection obtained here is exact. Its Taylor coefficient is

$$[z^{p-1}u^{r-1}]\{A(z, u) + A(-z, -u)\} = (1 + (-1)^{p+r})Q_{p,r}.$$

Since $S_6 = Q_{6,1}$ and $S_8 = Q_{8,1}$ have total weights 7 and 9, their coefficients vanish in this projection. The same is true of the weight-5 S_4 value: its known proof uses additional functional information. Further differentiation of the same even projection does not recover its absent Taylor coefficients. A successful extension must control the complementary function $A(z, u) - A(-z, -u)$ or produce an independent relation among the required mixed-color periods.

The separating functional already exhibited in the manuscript rules out a particular restricted row-span certificate for the S_6 candidate. It does not rule out the identity itself. Similarly, a formal quotient dimension does not prove arithmetic independence of the numerical periods. These distinctions are retained throughout this continuation.

6.3 Audit findings and concrete integration corrections

No new false mathematical statement was found in the sampled pinned Gaussian claims or the incoming correlation identities used here. Several concrete reconciliations are nevertheless needed when the packages are integrated.

1. **Reconcile the open-question lists.** The Closure report's questions about undilated parameter derivatives and associative convolution are answered in the current Shifted Hurwitz and Convolution packages. Their entries should point to those results. The integer-dilation questions should point to the present scale and product theorems, with disjointness and ambient normalization stated.
2. **Keep the two dilation operations distinct.** The replacement $X_{q,n,r} = U_q T_{n,r}$ would omit (5.4). A simple exact diagnostic is

$$\langle U_q G_0, 1 \rangle = 0, \quad \langle X_{q,0,0}, 1 \rangle = -\log q.$$

Thus they already differ at zero Stieltjes index and zero derivative order. This corrects a possible naive extension of the incoming formulas; it does not assert that the supplied undilated theorem made that error.

3. **State the cutoff variable in derivative identities.** The valid undilated assertion $h_{r,n} = 0$ for $n \geq r$ cannot be copied unchanged into a theorem about a fixed ambient scale. For example,

$$DX_{q,1,0} = qX_{q,1,1} - \frac{\log q}{q} \Delta'_q.$$

A separate constant term in each local variable has the canonical normalization used here. Rescaling those variables before one common constant-term extraction changes it.

4. **Retain resonance values before simplifying factors.** At integer spectral parameters, use the analytic germ (3.11). The equality $F_0(1) = 1$ is responsible for the $-1/4$ and $-3/4$ higher-pole examples. Dropping a factor that vanishes next to a zeta pole is not a valid simplification.
5. **Preserve the existing conjecture and error history.** The present S_8 candidate is different from the earlier vector already rejected in `10-discovery.tex`. The mixed-color symmetry counterexample is already recorded in `04-depth.tex`. These are existing

corrections, not new discoveries of this audit. The generalized-harmonic family $Q_{p,r}$ should also retain its own notation, since the manuscript's $S_{p,r}$ uses elementary symmetric harmonic sums.

6.4 Proposed research programme

The following questions seek exact identities or structural reductions. They are research questions, not conjectures inferred from a few numerical samples. The first three are the most immediate next steps.

- Q1. The missing Gaussian projection.** Find a functional equation or convergent iterated-integral relation that determines the weight-7 and weight-9 coefficients of $A(z, u) - A(-z, -u)$. A concrete success criterion is an exact zero certificate for (6.1), followed by the current S_8 vector. The known S_4 octahedral seed suggests testing genuinely new functional relations before enlarging a numerically fitted coefficient basket.
- Q2. Colliding unequal dilations.** Extend the ambient-coordinate product theorem to $Pa \in \mathbb{Z}$. There the partition separating the two singular grids fails, and the product contains higher singular powers at the common points. Derive the full local subtraction polynomial and compare its finite value with the collision expansion of (5.20). The already known undilated coincident closure is a boundary case and a normalization check, not a solution of every unequal-grid configuration.
- Q3. Nonlinear changes of local coordinate.** Replace qt near a singularity by an analytic map $f(t) = qt + c_2t^2 + \dots$, $q > 0$. Determine the finite combination of delta derivatives that converts pullback into ambient finite part for every $\gamma_n^{(r)}$. The linear coefficient must recover $c_{n,r}(\log q)$; higher coefficients should be computable from finite Taylor and Bell-polynomial algebra. The goal is a proved composition law for the full local correction.
- Q4. Three translated or dilated factors.** For three pairwise disjoint singular grids, determine the spectral generating kernel of the product of three Hurwitz or Stieltjes functions. Fourier orthogonality leaves a two-dimensional frequency constraint. Identify which double Hurwitz sums or multiple polylogarithms survive, and derive the first nontrivial exact Stieltjes coefficient formulas. Do not assume the bilinear single-Stieltjes closure persists at degree three.
- Q5. Complete primitive calculus for dilated products.** Insert the finite closure into the anchored Hurwitz-jet primitive ladder to construct explicit primitives in the shift a on every noncollision interval. Then determine the matching constants when crossing a collision point in a specified finite-part convention. This would connect local exact integration with a global periodic distribution statement.
- Q6. Character-weighted dilation and multiplication.** Replace the unweighted trace by Dirichlet-character weights on the finite covering. Determine how the local correction interacts with the character's zero mode, and express the resulting correlations in Dirichlet L -jets. The residue-class trace polynomial (5.19) supplies the unweighted test case.
- Q7. Rational-shift harmonic identities beyond a reflection.** Combine the two-shift theorem with recurrences in its inner and outer shifts. Determine which finite combinations of individual shifted harmonic sums admit depth-one cyclotomic expressions, rather than only their reflected pairs. A useful first classification fixes small denominators in u and z and records exact functional certificates.

- Q8. Mellin kernels with noninteger denominator order.** Replace N in $(1+x)^{-N}$ by a complex parameter. Find a spectral representation that specializes to the finite polynomial operator at positive integers, and determine exactly when a finite ordinary-polylogarithm closure survives. Hypergeometric or generalized Lerch functions are natural intermediate objects, but their branch and resonance conventions must be derived from the convergent integral.
- Q9. Complex poles and contour rotation.** Extend Corollary 2.7 to rational kernels with nonreal poles. Track the residue terms when a rotated integration ray crosses a polylogarithm cut. The target is an exact continuation formula that reduces to the positive-real scaling law when every pole lies on the negative real axis.
- Q10. Products of polylogarithms under the Mellin kernel.** Study

$$\int_0^\infty \frac{x^{a-1} \operatorname{Li}_s(-zx) \operatorname{Li}_t(-wx)}{(1+x)^N} dx$$

where ordinary convergence holds. Decide whether integer Mellin parameters lead to a finite multiple-polylogarithm algebra, and identify the minimal depth forced by two independent arguments. The one-factor integer table gives exact boundary cases and a recurrence in N .

- Q11. Compatibility of the two harmonic coefficient calculi.** Both cancelled zeta poles and dilation corrections involve $\prod_j (1 - u/j)$, whereas repeated anchored primitives involve its reciprocal. Formulate and prove a finite algebra connecting these operations, including derivative-primitive commutators and their integration polynomials. This could reduce duplicated coefficient calculations across the manuscript.
- Q12. Exact certificates and formal verification.** Encode the polynomial recurrences, the scale composition law, and the integer moment table as exact certificates with stable normalization metadata. A proof-assistant development could then isolate the finite algebra from the analytic tasks of dominated parameter differentiation, meromorphic continuation, and distributional pullback. Numerical agreement alone should not change theorem status.

These questions are constrained by the proved formulas: each has a specified base case, normalization, or missing component to resolve. They also preserve the main distinction of this article: ordinary polylogarithmic integrals, finite harmonic reductions, and normalized distribution products are connected by exact operations, but are not interchangeable without their hypotheses.

A Reproduction, verification, and source provenance

A.1 What is proved, and what the computations check

The arbitrary-order identities are established by the analytic proofs in the main text. The exact program checks finite algebra in specified ranges. The numerical programs compare independent expressions and detect normalization, coefficient and branch mistakes. No numerical residual is treated as a proof of equality, and no proof-assistant formalization is claimed.

The principal exact suite reports 742 successful assertions:

Exact assertion family	Count
Polynomial recurrence, $N = 1, \dots, 8$	8
Initial-term annihilation, $N = 1, \dots, 8$	28
Integer coefficient determinant, $N = 1, \dots, 8$	8
Independent rational-polylogarithm logarithmic moments	525
Two scale-composition identities, $0 \leq r \leq 5, 0 \leq n \leq 6$	84
Contact-polynomial comparison in the same range	42
Undilated closure symmetry, top orders, and explicit base rows	47
Total	742

The 525 moment checks cover $1 \leq N \leq 6$, every $0 \leq m < N$, $s = 0, -1, \dots, -4$, and logarithmic powers $0 \leq h \leq 4$. One side uses (2.19), with every cancelled zeta pole evaluated by (3.11). The other side starts from the rational negative-order polylogarithm, integrates its terms by the beta function, and differentiates that beta integral. Complete Bell polynomials in polygamma differences evaluate these independent derivatives. Thus this comparison does not merely call the same coefficient formula twice.

The 15 undilated coefficient rows through $m+n \leq 4$ retain formal symbols $Z_j = \zeta(j)$. Their symmetry and missing-order assertions are finite algebra checks of a previously established input theorem. They do not assume algebraic independence of the zeta values. The symbolic jet script separately confirms the three displayed $s = -1$ resonance values and the third-denominator derivative example.

A.2 Recorded numerical comparisons

There are 77 recorded numerical comparisons in the package. Working precision and observed residuals are reported separately: neither is an interval certificate.

Comparison	Cases	Digits	Largest recorded discrepancy
Integer Mellin rows, including complex spectral order and $z > 1$	17	28	scaled, $< 10^{-26}$
Noninteger complex Mellin master	4	30	scaled, $< 2.6 \cdot 10^{-25}$
Unshifted harmonic parity against quadrature	25	60	absolute, $< 3 \cdot 10^{-58}$
Bivariate reflection, mixed derivatives, and quarter shift	7	180	absolute, $< 4 \cdot 10^{-136}$
Unequal-dilation polygamma products	9	42	scaled, $< 3.5 \cdot 10^{-39}$
Positive-index Stieltjes products	3	28	scaled, $< 9 \cdot 10^{-29}$
Anchored primitives and jet examples	11	42	absolute, $< 3 \cdot 10^{-43}$
The separate π^2 finite-part example	1	42	absolute, $< 3.9 \cdot 10^{-40}$

Each program specifies its normalization. The scaled error divides the absolute discrepancy by one of the two recorded scales

$$\max(1, |\text{lhs}|, |\text{rhs}|), \quad \max(1, |\text{rhs}|).$$

The Mellin quadrature uses $x = t/(1-t)$ and evaluates the polylogarithm on the negative ray. The closed side uses the finite integer formula or the Lerch master expression. Three general-master cases have $1 < \Re a < N$, explicitly testing the safe parameter $N - a$ beyond the initial one-pole strip.

The harmonic tests use two representations: ordinary real quadrature for the scalar family, and a 450-term Euler transformation of the defining alternating sums for the shifted family. The latter includes a complex pair and a pair outside the initial small-parameter proof neighborhood.

The polygamma code integrates the explicitly subtracted ordinary integrands in (5.30); the other side uses the Hurwitz-zeta formula. The positive-index Stieltjes code similarly subtracts the exact local logarithmic singularity. To avoid subtraction loss extremely close to an endpoint, that diagnostic uses four Taylor terms for the smooth difference when $t < 10^{-8}$. This is a numerical stabilization, not a certified remainder argument. The analytic proof uses the exact smooth difference. Exceptional primitive values are checked using a finite Cauchy mean of the complete closed expression against a direct anchored integral, with no individual pole evaluated.

A.3 Commands and package contents

The tested numerical environment is Python 3.12.14, mpmath 1.3.0, and SymPy 1.14.0. The exact versions are recorded in the results. The source uses standard pdfLaTeX packages; no custom font or network download is needed once its dependencies are installed.

From the extracted package directory, the principal commands are

```
python -m pip install -r requirements.txt
make pdf
make check-exact
make check-numeric
```

The numerical target intentionally runs all recorded families and can take substantially longer than compilation or the exact suite. Each program can also be run individually. Outputs go to `results/`.

Path	Content
Main <code>.tex</code> and <code>.pdf</code>	Complete standalone article.
<code>sections/</code> ,	Modular mathematical source and bibliography.
<code>references.tex</code>	
<code>code/</code>	Exact coefficient engine and independent numerical programs; no integer-relation search is used.
<code>results/</code>	Exact coefficient tables, recorded comparisons, verification summary, and build audit.
<code>data/</code>	Pinned source manifest with file sizes and SHA-256 hashes.
<code>integration/</code>	Status and source audits, integration guidance, and a generated mathematical fragment.
SHA256SUMS	Hashes of the final packaged artifacts.

A.4 Snapshot and integration scope

The source baseline is commit `fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed` of Vladimir Reshetnikov’s ProveIt repository. The manifest covers 86 retrieved files from the requested manuscript and incoming directories, including all five incoming ZIP archives. It identifies the source snapshot; it does not imply that every retrieved statement was audited in full. The selected proof and conjecture audits are documented separately.

The standalone article was compiled and its rendered pages inspected. The integration fragment is supplied to make a later manuscript merge concrete, with namespaced labels and an explicit preamble dependency list. It was not compiled inside the full canonical manuscript. No repository branch, source file, issue, or pull request was changed by this delivery. Integration should preserve the existing proven and conjectural statuses and remove duplicate presentations of the undilated input theory.

References

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- [2] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*, especially the series, continuation, and Fermi-integral formulas. <https://dlmf.nist.gov/25.12>. Accessed 10 October 2026.
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- [8] ProveIt incoming research, *Resonant Lerch Limits and Shifted Gamma Correlations: Uniform Herglotz asymptotics and nonseparable distribution jets*, 10 October 2026. Archive `ProveIt_Resonance_Correlations_2026-10-10.zip` in `docs/incoming` at the snapshot of [1].
- [9] ProveIt incoming research, *Shifted Hurwitz-Zeta Correlations: Exact Stieltjes Closure, Polygamma Contact Terms, and Log-Gamma Antiderivatives*, 10 October 2026. Archive `ProveIt_Shifted_Hurwitz_Jets_2026-10-10.zip` in `docs/incoming` at the snapshot of [1].
- [10] ProveIt incoming research, *A Convolution Algebra of Stieltjes Functions: Finite-part identities, polygamma contact terms, log-gamma convolution powers, and polylogarithmic correlations*, 10 October 2026. Archive `ProveIt_Stieltjes_Convolution_2026-10-10.zip` in `docs/incoming` at the snapshot of [1].
- [11] ProveIt incoming research, *Bilinear Closure of Generalized Stieltjes Correlations: Gamma-ratio kernels, polylogarithmic jets, exact collision expansions and antiderivatives*, 10 October 2026. Archive `ProveIt_Stieltjes_Correlation_Closure_2026-10-10.zip` in `docs/incoming` at the snapshot of [1].
- [12] ProveIt incoming research, *A Convolution Calculus for Stieltjes Functions: Finite parts, polylogarithm jets, polygamma corrections, and sharp log-Gamma correlations*, 10 October 2026. Archive `Stieltjes_Convolution_Calculus_2026-10-10.zip` in `docs/incoming` at the snapshot of [1].